HSPT Exam Simulator

Full-Length Practice Exam 13

Complete printable test with 298 multiple-choice questions, answer key, and detailed explanations.

Section	Questions	Time
Verbal Skills	60	20 minutes
Quantitative Skills	52	35 minutes
Reading Comprehension	62	35 minutes
Mathematics	64	45 minutes
Language	60	30 minutes

General Directions

Work through each section within the recommended time. Choose the best answer for every question. Mark one answer (A-D) for each item. Use the answer key and explanations only after completing the test.

Recommended total testing time: 165 minutes. Calculators should be used only when permitted by the rules followed for your HSPT preparation.

Section 1: Verbal Skills

60 questions | Recommended time: 20 minutes

1. Which phrase is closest in meaning to debris?

A) a complete structure

B) a detailed plan

C) scattered remains

D) an untouched supply

2. Which word is closest in meaning to tedious, as in a tedious task?

A) hazardous

B) profitable

C) wearisome

D) unpredictable

3. Which phrase best expresses ambivalence about a choice?

A) complete absence of interest

B) unquestioning acceptance of advice

C) simultaneous conflicting feelings

D) inability to remember the alternatives

4. Which word is most nearly opposite in meaning to reticent, describing a person reluctant to speak?

A) guarded

B) reserved

C) tight-lipped

D) talkative

5. Which word is most nearly opposite in meaning to external, as in an external coating?

A) internal

B) outer

C) exterior

D) outside

6. Which word is most nearly opposite in meaning to magnanimous toward someone who has caused an offense?

A) vindictive

B) forgiving

C) charitable

D) noble

7. Nocturnal is to night as diurnal is to ____.

A) twilight

B) midnight

C) daytime

D) winter

8. An embargo restricts trade. In a similar relationship, a quarantine restricts ____.

A) recovery

B) nutrition

C) diagnosis

D) movement

9. Alphabetical order is to a dictionary as chronological order is to ____.

A) a compass

B) a timeline

C) a portrait

D) a balance

10. Which word names a formal agreement rather than a kind of conversation?

A) treaty

B) interview

C) banter

D) dialogue

11. In which pair does capitalizing the first letter create a different word meaning, rather than merely place the word at the start of a sentence?

A) polish; Polish

B) repair; Repair

C) collect; Collect

D) observe; Observe

12. Which pair consists of two words that imitate the sounds they name?

A) melody, harmony

B) hiss, clang

C) volume, resonance

D) accent, intonation

13. Start with ABC. A legal move replaces adjacent AB with BA or adjacent BC with CB. Any number of legal moves, including zero, is allowed. Which arrangement can never be reached?

A) ABC

B) CAB

C) BAC

D) ACB

14. A design is ruled out if another design is both lighter and cheaper. A is lighter but more expensive than B. C is lighter and cheaper than B. Nothing compares A with C. Which design must be ruled out?

A) A

B) B

C) C

D) Every design

15. Someone rejects a claim about a satellite's orbit solely because its author has messy handwriting. What is the flaw in this reasoning?

A) It proves the orbit calculation uses the wrong units.

B) It establishes that every handwritten claim is false.

C) It checks the claim against independent observations.

D) It substitutes an unrelated personal characteristic for evidence about the claim.

16. Which word is closest in meaning to discrepancy, as in a discrepancy between two accounts?

A) agreement

B) exaggeration

C) explanation

D) mismatch

17. Which phrase best expresses savor in the sentence "He paused to savor the final bite"?

A) save for later

B) enjoy fully

C) examine for flaws

D) swallow hurriedly

18. Which phrase is closest in meaning to rapport between two people?

A) a relationship of mutual understanding

B) a requirement to agree on everything

C) a formal division of responsibilities

D) a competition for the same position

19. Which phrase is most nearly opposite in meaning to portable, describing equipment?

A) light in weight

B) easily damaged

C) fixed in place

D) adjustable in height

20. Which phrase is most nearly opposite in meaning to piecemeal, as in introducing changes piecemeal?

A) little by little

B) all at once

C) without a plan

D) with great reluctance

21. Which phrase is most nearly opposite in meaning to imminent, describing an event?

A) far off in time

B) unlikely to occur

C) difficult to predict

D) brief in duration

22. A pseudonym substitutes for an author's public name as a stage name substitutes for a performer's ____.

A) audience

B) public name

C) costume

D) place of birth

23. A lens brings light to a focus. In a similar relationship, a funnel brings a flowing liquid toward ____.

A) many separate sources

B) one narrow outlet

C) a frozen surface

D) an expanding mist

24. A question is interrogative as a command is ____.

A) declarative

B) conditional

C) descriptive

D) imperative

25. Which word names a boundary marking where something ends, rather than a feature projecting beyond a surface?

A) ledge

B) spur

C) bulge

D) limit

26. Which pair contains words that share pronunciation but differ in both spelling and meaning?

A) lead; lead

B) bright; brilliant

C) seam; seem

D) rough; tough

27. In which pair does the second word describe a position dependent on the observer, rather than a fixed compass direction?

A) east; west

B) north; south

C) left; right

D) northeast; southwest

28. One dial controls volume and another controls brightness. Starting with both low, either dial may be raised without changing the other. Which claim contradicts that independence?

A) Raising volume always raises brightness.

B) Either dial can be adjusted first.

C) Brightness may be high while volume is low.

D) Both settings can be high together.

29. A sign says, "Only permit holders may enter." Dev has a permit. What follows from the sign alone?

A) Dev must be allowed to enter regardless of other rules.

B) Anyone without a permit may enter when Dev enters.

C) Having a permit is required for entry, but it does not by itself guarantee entry.

D) Everyone who has a permit must enter.

30. A critic says, "Design P deserves first place because it is the best design." Asked how he knows it is best, he replies, "Because it deserves first place." What is the flaw?

A) The critic compares two measurements using different units.

B) The critic shows that no ranking can ever use evidence.

C) The critic supplies a test result that contradicts his conclusion.

D) Each claim is offered as support for the other without an independent reason.

31. Which phrase is closest in meaning to quota, as in a worker's production quota?

A) required target amount

B) unlimited supply

C) unexpected reward

D) average price

32. Which phrase best expresses levity during a conversation?

A) careful secrecy

B) deep resentment

C) lighthearted humor

D) strict obedience

33. Which phrase best explains a musician's repertoire?

A) the equipment used to record a performance

B) the audience expected at a concert

C) the collection of works the musician can perform

D) the instructions for repairing an instrument

34. Which word is most nearly opposite in meaning to domestic, describing goods produced within a country?

A) household

B) handmade

C) inexpensive

D) imported

35. Which phrase is most nearly opposite in meaning to squalid, describing living conditions?

A) small but affordable

B) distant and isolated

C) clean and well-kept

D) crowded but lively

36. Which word is most nearly opposite in meaning to innocuous, describing a substance?

A) unfamiliar

B) colorless

C) ineffective

D) harmful

37. Interest is calculated on loan principal as a percentage-based sales commission is calculated on ____.

A) the number of hours in a day

B) the value of sales

C) the age of the salesperson

D) the balance of an unrelated loan

38. Hatching is to an eggshell as sprouting is to a seed coat. What relationship do both pairs express?

A) A new organism emerges through its enclosing covering.

B) A covering grows thicker as the organism becomes dormant.

C) An organism returns inside the covering it previously left.

D) A covering supplies the organism with its permanent outer skeleton.

39. A bookmark marks a reader's current place in a text as a cursor marks a user's current ____.

A) position on a screen

B) password for an account

C) connection speed

D) copy of a printed page

40. Consider fuel fed into an engine, exhaust released by that engine, ash left by a fire, and sawdust left by sawing. Which substance is an INPUT to its process as described?

A) fuel

B) ash

C) sawdust

D) exhaust

41. Which pair contains two blended words formed from parts of longer words, rather than two complete words joined together?

A) notebook; fireplace

B) raincoat; sunrise

C) toothbrush; snowball

D) smog; brunch

42. In which pair does EACH word begin with a consonant letter that is silent in standard American pronunciation?

A) lamb; debt

B) rhyme; wrist

C) knot; grip

D) knock; write

43. Two articles repeat the same unverified claim. Both authors obtained it from the same announcement. Why does agreement between the articles fail to provide two independent confirmations?

A) The agreement proves that the claim is false.

B) Every announcement is less reliable than every article.

C) The matching claims share one underlying source.

D) Different authors are incapable of reaching the same correct conclusion.

44. A rule states, "Every parcel on Cart R has a red label." Which situation is consistent with the rule?

A) Cart R contains a parcel with no label.

B) Cart R contains no parcels.

C) Cart R contains a parcel whose only label is blue.

D) Cart R contains a parcel with a green label and no red label.

45. To know a claim, the claim must be true; a belief may be mistaken. Ivo knows that Rae believes the blue box is empty. What must be true?

A) The blue box is empty.

B) Rae believes that the blue box is empty.

C) Rae knows that the blue box is empty.

D) Ivo believes that the blue box is not empty.

46. Which phrase is closest in meaning to haven, as in a haven during a storm?

A) a warning of danger

B) a period of travel

C) a place of safety

D) a source of noise

47. Which word is closest in meaning to lapse in the phrase "a lapse in attention"?

A) failure

B) improvement

C) requirement

D) exchange

48. Which phrase best explains a harbinger of change?

A) a person who prevents every change

B) a record written after a change

C) a sign that change is approaching

D) a rule requiring changes to occur regularly

49. Which word is most nearly opposite in meaning to urban, describing an area?

A) crowded

B) coastal

C) industrial

D) rural

50. Which phrase is most nearly opposite in meaning to lopsided, describing a display?

A) brightly colored

B) evenly balanced

C) unusually large

D) carefully hidden

51. Which phrase is most nearly opposite in meaning to infallible, describing a judge?

A) unable to reach a decision

B) capable of making mistakes

C) unwilling to hear advice

D) certain to make every decision incorrectly

52. A hinge lets a door swing, while a latch can hold it shut. Similarly, an open valve lets liquid pass, while a ____ is used to block the passage.

A) plug

B) pump

C) nozzle

D) gauge

53. A monologue has one speaker as a monopoly has one ____.

A) buyer

B) product

C) advertisement

D) seller

54. A footprint is a trace left by the passage of a foot. In the same relationship, a wake on the water is a trace left by a ____.

A) cloud

B) candle

C) bell

D) boat

55. Which word names a step rather than a shaking movement?

A) tremor

B) shudder

C) quiver

D) stride

56. Which group consists entirely of words naming visible light or shine, rather than colors, coloring substances, or lack of light?

A) crimson, violet, blue

B) ink, dye, pigment

C) glare, glow, gleam

D) shadow, darkness, gloom

57. A label is self-describing if its meaning accurately describes its own printed form. Which label below is self-describing?

A) unhyphenated

B) uppercase

C) hyphenated

D) THREE WORDS

58. A reviewer says, "Either accept every suggested change or refuse all help." A writer can accept useful suggestions while rejecting others. What is wrong with the reviewer's argument?

A) It proves that every suggestion is equally useful.

B) It shows that accepting help requires no decisions.

C) It excludes a selective option that is available.

D) It establishes that rejecting one change makes every other change impossible.

59. A walking route consists of two straight segments. Jo concludes that the whole route must be one straight line. Which fact exposes the flaw?

A) Each segment has a beginning and an end.

B) The two segments may meet at an angle.

C) A person may walk slowly along either segment.

D) The route can be described using words.

60. Ari and Bea each wear a red or blue badge and can see only the other person's badge. Both reason correctly and know at least one badge is red. Ari truthfully says, "I cannot determine my own color." What must Bea conclude?

A) Ari's badge is blue.

B) Bea's badge is red.

C) Ari's badge is red.

D) Both badges are blue.

Section 2: Quantitative Skills

52 questions | Recommended time: 35 minutes

61. Term n is the least common multiple of all positive integers from 1 through n. Which term follows?
1, 2, 6, 12, 60, 60, 420, ____

A) 420

B) 1,260

C) 3,360

D) 840

62. Start with 0. If the last number had not appeared earlier, write 0 next. Otherwise, write the distance in positions back to its most recent earlier appearance. What comes next?
0, 0, 1, 0, 2, 0, 2, 2, 1, ____

A) 1

B) 6

C) 3

D) 8

63. To get each next term, add 2 to the current term and then take the reciprocal. What comes next?
1, 1/3, 3/7, 7/17, ____

A) 17/31

B) 7/19

C) 17/41

D) 41/17

64. At each step, arrange the four digits in descending order and in ascending order, retaining any leading zero, then subtract the smaller arrangement from the larger. What comes next?
3524, 3087, 8352, ____

A) 6174

B) 6176

C) 8532

D) 2358

65. A positive integer d divides N when N = d × k for some integer k. Which positive integers divide 0?

A) Only 1

B) Only even positive integers

C) Only positive powers of 10

D) Every positive integer

66. One of 27 identical-looking samples is lighter than all the others, which have equal mass. A balance compares equal-sized groups. Which first weighing leaves the fewest possible candidates in the worst case?

A) 8 samples against 8 samples

B) 10 samples against 10 samples

C) 9 samples against 9 samples

D) 13 samples against 13 samples

67. How many two-digit positive integers contain exactly one digit 5?

A) 18

B) 17

C) 19

D) 9

68. Six packets are placed in five bins, with every packet placed in one bin.
Quantity A: the greatest number of packets in any single bin.
Quantity B: 1.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

69. A list contains four different real numbers.
Quantity A: the median of the list.
Quantity B: the mean of its two greatest numbers.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

70. A data set contains at least two numbers. Add 7 to every number.
Quantity A: the new range.
Quantity B: the original range.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

71. A central angle and an inscribed angle intercept the same nonzero arc of a circle.
Quantity A: the central angle.
Quantity B: the inscribed angle.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

72. In triangle ABC, AD bisects angle BAC and D lies on BC. An internal angle bisector divides the opposite side in the ratio of the adjacent sides. AB = 5 and AC = 8.
Quantity A: BD.
Quantity B: DC.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

73. Two closed right circular cylinders have equal total surface areas, including both ends. Their positive radii and heights are otherwise unspecified.
Quantity A: the first cylinder's volume.
Quantity B: the second cylinder's volume.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

74. The entries count the 1s in the base-two representations shown. What is the next entry?
Base two: 0, 1, 10, 11, 100, 101, 110, 111, 1000
Entries: 0, 1, 1, 2, 1, 2, 2, 3, ____

A) 0

B) 4

C) 1

D) 8

75. Fractions are listed in groups with numerator plus denominator equal to 2, then 3, then 4, and so on. Within each group, numerators increase by 1. Fractions are not reduced. What comes next?
1/1, 1/2, 2/1, 1/3, 2/2, 3/1, 1/4, 2/3, ____

A) 3/2

B) 3/4

C) 4/1

D) 2/2

76. The list contains, in increasing order, all positive integers written using only the digits 2 and 7. What comes next?
2, 7, 22, 27, 72, 77, ____

A) 227

B) 272

C) 222

D) 777

77. A proposed proof starts with a = b = 4. It correctly obtains (a − b)(a + b) = b(a − b), then cancels a − b and claims a + b = b. Why is that cancellation invalid?

A) It subtracts unequal values.

B) It uses a negative value for b.

C) It changes the order of addition.

D) It divides by zero.

78. A four-digit code passes a check when its first digit plus three times its second digit plus its third digit plus three times its fourth digit is divisible by 10. The code 2754 passes. Which exchange also produces a passing code?

A) Exchange the first and second digits.

B) Exchange the second and third digits.

C) Exchange the first and fourth digits.

D) Exchange the third and fourth digits.

79. Five different positive integers have sum 40. What is the greatest possible value of the smallest integer?

A) 5

B) 8

C) 30

D) 6

80. A positive integer leaves remainder 4 when divided by 7. Which list gives ALL its possible remainders when divided by 14?

A) 4 only

B) 7 and 11

C) 4 and 11

D) 4, 7, and 11

81. For a noninteger real number x, let floor(x) mean the greatest integer less than or equal to x.
Quantity A: x.
Quantity B: floor(x).

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

82. Positive numbers a and b satisfy a + b = 1.
Quantity A: $a^2 + b^2$.
Quantity B: 1.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

83. Events A and B each have probability 1/2. No information about their dependence is given.
Quantity A: the probability that both occur.
Quantity B: 1/4.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

84. A square is enlarged by a factor of 2 and then shifted right. An identical square is shifted right and then enlarged by a factor of 2.
Quantity A: the first resulting area.
Quantity B: the second resulting area.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

85. A regular hexagon has side length s > 0. Its center and vertices form six equilateral triangles.
Quantity A: the straight distance between opposite vertices.
Quantity B: the length of three consecutive sides.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

86. A rhombus has no right angles. A square has the same perimeter as the rhombus.
Quantity A: the rhombus's area.
Quantity B: the square's area.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

87. Each term describes the consecutive runs of digits in the preceding term: 21 means "one 2, one 1," producing 1211. What comes next?
1, 11, 21, 1211, 111221, ____

A) 311221

B) 132211

C) 111222

D) 312211

88. Start with 1. Each next term is the smallest larger integer keeping ALL unordered pair sums different, including a term added to itself. What comes next?
1, 2, 4, 8, ____

A) 13

B) 9

C) 11

D) 12

89. For denominators 2, 3, 4, and so on, count the positive proper fractions already in lowest terms. For denominator 4, count 1/4 and 3/4 but not 2/4. What is the next count?
Denominators: 2, 3, 4, 5, 6, 7, 8, 9, 10
Counts: 1, 2, 2, 4, 2, 6, 4, 6, ____

A) 3

B) 5

C) 8

D) 4

90. Term n is the remainder when n^2 is divided by 7. What comes next?
1, 4, 2, 2, 4, 1, ____

A) 2

B) 0

C) 6

D) 7

91. For any nonzero number k, what is k raised to the power 0?

A) 0

B) k

C) undefined

D) 1

92. A length u measures both 2 1/4 and 3 3/8 exactly if each length is a positive whole-number multiple of u. What is the greatest possible u?

A) 3/8

B) 3/4

C) 2 1/4

D) 1 1/8

93. Three positive integers have sum 12. What is the greatest possible product?

A) 60

B) 64

C) 72

D) 144

94. Use the names A, B, and C.
Quantity A: the number of ordered two-name lists with different names.
Quantity B: the number of two-name groups, where order does not matter.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

95. The integer n is positive.
Quantity A: (2/3) raised to n.
Quantity B: (3/4) raised to n.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

96. Positive numbers a and b satisfy ab = 1.
Quantity A: 1/(1 + a) + 1/(1 + b).
Quantity B: 1.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

97. A circle has positive radius.
Quantity A: its area divided by its diameter.
Quantity B: one-fourth of its circumference.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

98. A right triangle has hypotenuse 10 units. M is the midpoint of that hypotenuse. In a right triangle, the hypotenuse midpoint is equidistant from all three vertices.
Quantity A: the distance from M to the right-angle vertex.
Quantity B: 5 units.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

99. A right circular cone and a right circular cylinder have the same positive height and equal volumes.
Quantity A: the cone's radius.
Quantity B: the cylinder's radius.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

100. Write the consecutive integers 98, 99, 100, 101, and so on without spaces, then read their digits one at a time. What comes next?
9, 8, 9, 9, 1, 0, 0, 1, 0, ____

A) 0

B) 2

C) 1

D) 9

101. Replace a triple (a, b, c) with (|a − b|, |b − c|, |c − a|), calculating all entries from the OLD triple. What comes next?
(2, 5, 9), (3, 4, 7), (1, 3, 4), (2, 1, 3), ____

A) (1, 1, 2)

B) (1, 2, 1)

C) (3, 4, 5)

D) (1, 2, 3)

102. Term n counts ways to write n as a sum of TWO OR MORE consecutive positive integers in increasing order. For n = 9, the ways are 4 + 5 and 2 + 3 + 4. What is the next count?
n: 9, 10, 11, 12, 13, 14, 15
Counts: 2, 1, 1, 1, 1, 1, ____

A) 3

B) 2

C) 4

D) 7

103. Express the ratio 2 meters : 50 centimeters using the SAME length unit on both sides, then reduce it.

A) 1:25

B) 1:4

C) 4:1

D) 40:1

104. A perfect number equals the sum of its positive divisors smaller than itself. Which number is perfect?

A) 28

B) 20

C) 24

D) 32

105. Without approximating either square root, find $\sqrt{6} \times \sqrt{24}$.

A) $\sqrt{30}$

B) 24

C) 144

D) 12

106. An object has mass 8 units. A balance also has weights of 2, 5, and 11 units, each used once. Which arrangement balances the object?

A) Put 2 with the object and 5 and 11 on the opposite pan.

B) Put 5 with the object and 2 and 11 on the opposite pan.

C) Put 11 with the object and 2 and 5 on the opposite pan.

D) Put all three weights on the pan opposite the object.

107. Quantity A: -0.003.
Quantity B: -0.03.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

108. Events A and B cannot occur together.
Quantity A: the probability that A or B occurs.
Quantity B: P(A) + P(B).

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

109. The number n is a positive integer. Count zero DIGITS in ordinary decimal notation, with no leading zeros.
Quantity A: the number of zero digits in n^2.
Quantity B: the number of zero digits in n.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

110. A straight line is tangent to a circle at T, and O is the center.
Quantity A: the angle between OT and the tangent line.
Quantity B: 75°.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

111. A trapezoid has two parallel bases of different positive lengths. Its midsegment joins the midpoints of the nonparallel sides and has length equal to half the SUM of the bases.
Quantity A: the midsegment length.
Quantity B: the longer base.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

112. AB is horizontal. A ray from A rises above line AB. A circle centered at B crosses this ray at two distinct points C and D, both beyond A. It is not stated which point is closer to A.
Quantity A: the area of triangle ABC.
Quantity B: the area of triangle ABD.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

Section 3: Reading Comprehension

62 questions | Recommended time: 35 minutes

Literary Passage: Everyone in the Picture

My task was simple: get everyone looking at the camera at the same time. I straightened shoulders, shifted a chair, and asked my cousin Selby to stop balancing on one foot. For a moment we were a shelf of upright books. My aunt set the camera to take its photograph after ten seconds and hurried into the row. Grandfather Rosh had left his straw hat on the table. Nobody was supposed to move now.

Then the garden door blew open. A draft lifted the hat, tipped it over the table's edge, and sent it rolling across the floor. Selby broke formation and caught it. Instead of putting it back, she placed it askew on her head. Grandfather began to laugh. My aunt bent forward to see what had happened. I turned to restore order, but Selby saluted me beneath the enormous brim, and I laughed too. The camera clicked while my eyes were nearly closed.

Later, we examined the photographs. In the earlier ones, every face was visible and every shoulder was straight. In the last, Grandfather's hand rested on Selby's shoulder, my aunt leaned into the group, and nobody looked directly at the lens.

"I failed," I said.

Grandfather chose the last photograph. "This one has all of us."

The others did too, of course. But I knew he was not counting faces.

113. Why does the camera take the last photograph while everyone is reacting to Selby?

A) Selby presses its button from across the room.

B) Grandfather secretly holds a remote control.

C) The garden door is connected to its shutter.

D) It has been set to take a photograph after ten seconds.

114. When Selby places the hat askew on her head, askew most nearly means ____.

A) securely

B) solemnly

C) invisibly

D) crookedly

115. Why does the narrator call the last photograph a failure even though all the family members appear in it?

A) It fails the original plan that everyone should look at the camera together.

B) It contains fewer people than the earlier photographs.

C) It leaves the garden door outside the frame.

D) It proves that the camera never completed its delay.

116. What does the comparison to "a shelf of upright books" emphasize?

A) The family's interest in reading during the gathering

B) The camera's inability to distinguish people from objects

C) The family's stiff, carefully arranged appearance before the interruption

D) The room's lack of enough chairs for the group

117. What does the final statement that Grandfather "was not counting faces" suggest about his choice?

A) He has forgotten how many family members attended.

B) He believes photographs should always hide people's faces.

C) He values the relationships and shared delight revealed by the photograph.

D) He has mistaken the number of people in the earlier images.

Informational Passage: Two Wheels in a Turn

Imagine a cart with two wheels of equal size on the same axle. On a straight path, both wheels travel equal distances. During a turn, however, the outside wheel must traverse a longer path than the inside wheel. Both complete the turn in the same time, so equal-sized wheels need different rotation rates if each is to roll without sliding.

Each full revolution of a wheel carries it forward by one circumference. The longer outside path therefore calls for more revolutions. A solid connection that forces both wheels to rotate at exactly the same rate prevents this difference. One wheel may then drag or slip rather than roll smoothly along its own path.

A differential is a mechanism that allows the two connected wheels to turn at different rates while receiving power. In this simplified account, its importance is not that it always makes one wheel faster. It allows a difference when the paths require one. During straight travel, equal rotation rates can still be appropriate.

The model depends on its assumptions. In particular, the wheels have equal circumferences. If their sizes differed, counting revolutions alone would no longer directly compare the distances traveled. The explanation links the shape of the route to wheel motion by keeping wheel size and elapsed time fixed, rather than treating a turn as though both wheels followed the same track.

118. Under the passage's equal-wheel assumption, why must the outside wheel make more revolutions during a turn?

A) It completes the turn in a longer time than the inside wheel.

B) It follows a longer path, while each revolution covers the same distance for either wheel.

C) Its circumference automatically becomes larger during every turn.

D) The inside wheel must stop moving whenever the cart changes direction.

119. What connection does the author primarily explain?

A) How unequal paths in a turn create a need for different rotation rates in equal-sized wheels

B) Why wheels must always rotate at different rates, even on a straight path

C) How a cart can move without any wheel turning

D) Why changing a wheel's color changes the distance it travels

120. Suppose the outside wheel is larger than the inside wheel. Why would its longer path alone no longer prove that it must make more revolutions?

A) Its larger circumference could cover more distance in each revolution.

B) A larger wheel must complete the turn at a different moment.

C) Different wheel sizes eliminate every difference in path length.

D) The number of revolutions would cease to have any connection with movement.

121. In the first paragraph, traverse most nearly means ____.

A) divide into pieces

B) look down upon

C) remove completely

D) travel along

122. What role does the final paragraph play in the explanation?

A) It provides a historical list of the first carts built in different countries.

B) It replaces the equal-wheel model with a claim that paths never differ.

C) It identifies assumptions that limit when revolution counts can directly compare distances.

D) It repeats instructions for painting the outside wheel.

Humanities Passage: An Imaginary Case, a Real Question

Two people want to share a coil of twine and a hammock. "Cut everything into equal halves," one proposes. The twine can become two usable lengths. Cutting the hammock down the middle, however, would not produce two working hammocks. The same physical procedure has different consequences for what the objects allow people to do.

This invented case is a thought experiment. It does not describe an actual disagreement or predict how particular neighbors will behave. Instead, it tests an attractive rule by making one distinction easy to examine. Dividing material equally is not always the same as sharing its benefit. Leaving other details out can isolate that distinction.

The missing details still matter if anyone wants a practical decision. The case says nothing about who already owns the objects. Nor does it explain whether both people need long pieces of twine or whether they could take turns using the hammock. Adding such facts might change the preferred arrangement without erasing the distinction the example revealed.

A useful thought experiment therefore makes its conditions clear and acknowledges what it leaves unsettled. Its simplicity can expose an assumption that ordinary discussion overlooks. That is a reason to use the imaginary case carefully, rather than either dismiss it for being invented or treat it as a complete answer to every real dispute. It is a question directed at a principle, not a verdict on unknown people.

123. What usable result does the proposed cutting produce for the twine?

A) two working hammocks

B) two shorter lengths

C) a single longer coil

D) a record of who owns it

124. What does the passage explain about the proper use of a thought experiment?

A) It provides a complete account of every participant's needs without additional facts.

B) It is useful only when its imaginary events have already happened to real people.

C) It proves that every physical object should always be divided into equal pieces.

D) It can test a principle by simplifying a case while leaving practical questions unresolved.

125. Suppose both people need twine longer than half the original coil. How would this added fact affect the discussion?

A) The distinction between dividing material and sharing benefit would disappear.

B) Equal cutting might no longer provide either person with the needed benefit.

C) The original case would become a factual report about identifiable neighbors.

D) Cutting the hammock would necessarily create two usable hammocks.

126. When the invented case can "isolate that distinction," isolate most nearly means ____.

A) consider separately

B) destroy permanently

C) measure with a ruler

D) prove beyond every possible doubt

127. What does the final contrast between a question and a verdict add to the passage?

A) It announces that no imagined example can reveal anything useful.

B) It frames the example as a way to investigate a rule rather than judge people whose circumstances are unknown.

C) It supplies the names and addresses missing from the opening story.

D) It changes the topic from sharing objects to conducting a criminal trial.

128. Which information does the author explicitly say the invented case omits?

A) whether the objects include a hammock

B) whether the proposed halves are equal

C) who already owns the objects

D) whether cutting the twine can create shorter lengths

Literary Passage: Not Yet

Neri laid the unopened envelope beside the brass weight on her desk. The roofers' workshop had promised to write about her request to become an apprentice. Now its answer was here, and she was polishing a weight that had not needed polishing yesterday.

"Not yet," she told the envelope.

She carried a cup to the kitchen. She returned a borrowed ruler. At the window she watched rain gather on a wire until one drop grew heavy enough to fall. The envelope waited without becoming either thinner or kinder.

Her brother came in looking for string. "Have you heard?"

"They've written."

He reached toward the desk, then stopped when she put her hand over the envelope. "You want me to leave?"

She had expected him to laugh at her waiting. His question gave her nothing to resist. She nodded, and he went out carrying the string.

Neri lifted the brass weight. Under it, the corner of the envelope had made a pale, clean triangle in the dust. She had spent the morning moving everything except the one thing that mattered.

Outside, another drop loosened from the wire. She slid a finger beneath the flap.

The room looked exactly as it had a moment before. She was glad of that. Whatever the letter said, she would read it in this room, standing on these boards, with the same rain at the window.

129. What answer is Neri expecting the workshop's letter to contain?

A) whether her brother has returned a ruler

B) whether she can become an apprentice

C) when the brass weight was manufactured

D) why rain is gathering on the wire

130. Which statement best expresses the story's central movement?

A) Neri discovers that the workshop has sent its letter to the wrong address.

B) Neri proves that her brother cannot be trusted with private information.

C) Neri begins to face an uncertain answer after delaying it through small tasks.

D) Neri decides that polishing brass matters more than learning a trade.

131. Why does her brother's question give Neri "nothing to resist"?

A) He tells her what the letter says before she opens it.

B) He offers her control over his presence instead of pressuring or teasing her.

C) He insists that she must explain her decision immediately.

D) He proves that she has already missed her opportunity.

132. What does the repeated idea of "not yet" help develop?

A) The tension between the answer's availability and Neri's delay in learning it

B) A dispute about whether the workshop has mailed any letter

C) A humorous account of a weight that keeps disappearing

D) A change from first-person to second-person narration

133. What is the main effect of the final emphasis on the unchanged room, boards, and rain?

A) It signals that the room has become part of a dream.

B) It gives Neri a sense of continuity while the letter's possible answer remains uncertain.

C) It proves that the workshop has accepted her request.

D) It suggests that the envelope contains a description of the weather.

Scientific Passage: Same Element, Different Structure

Graphite and diamond can both consist of carbon, yet they behave very differently. Graphite can leave a dark mark when drawn across paper. Diamond is extremely hard. The contrast cannot be explained simply by saying that one contains carbon and the other does not.

The arrangement of the carbon atoms matters. In graphite, atoms are strongly bonded within sheets. The attraction between those sheets is much weaker, allowing layers to slide and small pieces to rub off. A pencil mark therefore does not show that every bond in graphite is weak. It reveals a difference between the connections within a layer and the connections between layers.

In diamond, each carbon atom is bonded to neighboring atoms in a three-dimensional network. There is no equivalent stack of easily sliding sheets. The network helps explain why diamond resists scratching. Hardness here describes resistance to scratching, not a guarantee that a material can never break.

These structures show why identifying an element is only part of describing a material. Two samples can have the same kind of atom while differing in the way those atoms are connected. To understand a property, a useful question is not only "What is it made of?" but also "How is it put together?" Neither question replaces the other; the two descriptions work together.

134. According to the passage, what allows graphite layers to slide?

A) The attraction between sheets is weaker than the bonds within a sheet.

B) Graphite contains no bonds between any of its atoms.

C) Each sheet is made from a different chemical element.

D) Every carbon atom has to leave the material before a sheet moves.

135. What does the graphite-and-diamond comparison chiefly demonstrate?

A) Knowing a material's element always reveals every one of its properties.

B) All materials made from carbon are equally easy to scratch.

C) A material's visible color determines the strength of all its bonds.

D) Atomic arrangement can help explain different properties in materials made from the same element.

136. A reader concludes, "Because graphite rubs off on paper, all its atomic connections must be weak." Which distinction challenges that conclusion?

A) Rubbing off can involve separation between sheets without requiring all bonds within each sheet to be weak.

B) A pencil mark can form only when graphite changes into diamond.

C) No strong connection can exist in a material that makes a visible mark.

D) The passage says graphite contains no carbon.

137. In the phrase "resists scratching," resists most nearly means ____.

A) measures

B) withstands

C) requests

D) imitates

138. How does the author organize the explanation after introducing the contrast?

A) List prices, rank sellers, then recommend a purchase

B) Describe a mining journey, introduce a traveler, then resolve a conflict

C) Explain graphite's layers, explain diamond's network, then draw a broader conclusion about material descriptions

D) Give only dictionary definitions without connecting them to properties

Humanities Passage: One Melody, Several Starting Times

When singers perform a melody in unison, they sing its corresponding notes together. A round uses a different plan: groups sing the same melody but begin at different times. One group may be several measures into the tune when the next group enters at its beginning.

The delay does not necessarily mean that one group sings more slowly. With a common pace, the groups can maintain their separation as they proceed. Because different portions sound simultaneously, the combined texture differs from the single line heard at the opening. A melody chosen for a round must work with its own overlapping portions; not every tune produces a satisfying combination when entrances are staggered.

This overlap distinguishes a round from a simple echo in which groups take turns repeating a phrase, with only one group singing at a time. Both involve repetition, but only the described round has different portions of the same melody sounding together. Repetition alone is therefore not enough to identify the arrangement.

A written plan can make the timing visible. Imagine identical horizontal rows of notes shifted to the right for later entrances. Reading one row follows a group's melody through time. Looking straight down the rows at a particular moment shows what the groups are singing together. The tune belongs to each row; the texture depends on how those rows align. Listening to a round involves hearing both at once.

139. What do the groups share in the round described?

A) the same starting instant

B) a requirement to sing at different speeds

C) a rule that only one group may sing at a time

D) the same melody

140. Which statement best captures the explanation of a round?

A) A melody becomes a round whenever one singer repeats its final note.

B) Groups must continually change speed to prevent any overlap.

C) Delayed entrances let one melody form overlapping musical parts.

D) Unison singing and rounds require identical entrance times.

141. Two groups repeat the same phrase, but the second always waits until the first has finished it. Why would repetition alone not establish that they are singing the described round?

A) The groups have necessarily chosen different melodies.

B) The second group must be singing at half the first group's speed.

C) The singers cannot be following a written plan.

D) The required overlap of different portions has not been shown.

142. In "the combined texture differs from the single line," texture refers to ____.

A) the roughness of the paper holding the notes

B) the speed of one group considered alone

C) the total length of the performance

D) the way the simultaneous musical parts combine

143. In the written plan, what does looking straight down the rows at one moment show?

A) what the groups are singing simultaneously

B) which group owns the written page

C) the total number of performances in a year

D) the order in which instruments were invented

144. What does the imagined written plan contribute to the final paragraph?

A) It supplies evidence that every melody works as a round.

B) It changes the definition of a round to require different tunes.

C) It represents the difference between one part over time and several parts at one time.

D) It replaces the discussion with instructions for building an instrument.

Literary Passage: Something for the Seal

At the harbor office, Beto was sent to fetch the seal from the customs clerk. It was his first morning carrying messages, and he had already delivered a parcel to the wrong door. This time, he intended to show foresight. He borrowed a pail and filled it with water. At the fish stall he bought two small fish with his own coins. The stallholder asked whether he expected company. Beto said he could not discuss official business.

The customs clerk looked up from a stack of papers. "You've come for the seal?"

"Yes. Will it fit in here?" Beto raised the pail.

The clerk considered him for a moment, then opened a drawer. From it she lifted a brass stamp scarcely longer than her thumb. She pressed it into a patch of soft wax on an envelope. Its impression showed a ship above three waves.

Beto studied the fish. The clerk studied the pail.

"Very considerate," she said, putting the stamp into his empty hand. "But this one has never asked for lunch."

On the way back, Beto carried the seal in his pocket and the pail at arm's length. The stallholder called, "Did your company arrive?"

"Official business," Beto answered. He was relieved that the man could not see into his pocket. He was less relieved that the fish were so plainly visible.

145. What does the stamp leave in the soft wax?

A) the names of two fish

B) a ship above three waves

C) a picture of the harbor office door

D) an outline of Beto's pail

146. Which statement best summarizes Beto's experience?

A) A clerk refuses to let Beto carry any official document.

B) A stallholder tricks Beto into buying a damaged stamp.

C) Beto knowingly buys fish as a gift for the customs clerk.

D) An effort to anticipate a task's needs becomes embarrassing because he misunderstands a key word.

147. Why does Beto repeat "Official business" to the stallholder on his return?

A) He has been ordered to keep the stamp's design secret.

B) He uses the phrase to avoid explaining the embarrassing reason for his purchase.

C) He believes that the clerk still needs him to catch an animal.

D) He intends to sell the stamp in exchange for more fish.

148. When the wax receives the stamp's "impression," impression means ____.

A) a mark left by pressing

B) a private opinion

C) an imitation of a voice

D) a brief introduction

149. What makes the clerk's remark that the seal has "never asked for lunch" humorous?

A) She reveals that the stamp is secretly alive.

B) She changes the meaning of the word fish halfway through the sentence.

C) She playfully treats the stamp as the animal Beto expected.

D) She announces that Beto must buy another meal.

Literary Passage: A Hand Above the Table

At my first tenants' meeting, the chair proposed moving every bicycle from the entrance hall to the basement. I raised my hand with the others. I wanted my first contribution to be agreement.

Then I pictured the steep basement stairs. My delivery tricycle had a heavy box between its front wheels. I could not carry it down those stairs. Slowly, I lowered my hand.

The chair had already begun writing "All agreed" on the draft minutes. Beside me, a man closed his notebook. I could let the meeting move on and ask someone afterward. Afterward was beginning to sound very attractive.

I raised my hand again.

"Another vote?" the chair asked.

"A question. Does the plan include cycles that people cannot carry downstairs?"

I described the box and the stairs. The man beside me opened his notebook again. He used a hand-powered cycle, he said, and had been wondering the same thing.

The chair drew a line through the words she had written. Under them she added, "Who can use the proposed space?"

Nobody had solved the parking problem. The entrance hall still needed clearing, and the basement still had stairs. But the man turned his notebook toward me, leaving a blank space between our two names. This time, when the chair asked who wanted to speak, I kept my hand up.

150. What makes the proposed basement space difficult for the narrator to use?

A) The basement has already been filled with delivery boxes.

B) The narrator cannot carry the heavy delivery tricycle down the stairs.

C) The chair has forbidden tricycles from entering the building.

D) The narrator has lost the key to the entrance hall.

151. Which development is central to the story?

A) The narrator persuades every tenant to keep all cycles in the entrance hall permanently.

B) The chair announces a finished parking solution before the meeting begins.

C) The tenants discover that no one needs the basement stairs.

D) The narrator overcomes reluctance to object and brings an overlooked practical difficulty into the discussion.

152. Why does the chair replace "All agreed" with a question about who can use the space?

A) The discussion reveals that apparent agreement had concealed an access difficulty affecting more than one person.

B) The chair has discovered that the narrator has already solved every parking difficulty.

C) The other tenants have all withdrawn their support in a second vote.

D) The narrator has demanded that the minutes contain no questions.

153. How does the repeated raised-hand gesture help develop the narrator's change?

A) It proves that the narrator has cast several votes on the same finished proposal.

B) It shows that the narrator has forgotten what the chair asked each time.

C) It shifts from joining a vote to requesting an opportunity to speak.

D) It establishes that raising a hand always means refusing a proposal.

154. Why does the story end with continued discussion rather than a final parking decision?

A) It shows that the narrator's question has made the stairs disappear.

B) It emphasizes the opening of a more informed conversation rather than claim that the practical problem is already solved.

C) It establishes that the chair has refused to hear any further concerns.

D) It confirms that the original proposal has become a permanent rule.

Informational Passage: A Price Set by Another Bid

A fictional token auction separates the amount that wins from the amount the winner pays. Each bidder secretly writes one bid. The highest bidder receives the object but pays the second-highest bid. Only the winner pays. The examples here have no tied bids.

Suppose Mila bids nine tokens, Ren bids seven, and Tess bids four. Mila wins and pays seven tokens. Her bid determines her ranking, but Ren's bid sets her price. If Mila had bid eleven while the others kept their bids, she would still win and still pay seven. Raising an already winning bid would not, in that example, raise the payment.

That does not make a bid meaningless or guarantee that higher bidding is harmless. Suppose Mila is willing to spend no more than six tokens for the object but bids nine anyway. Against Ren's seven, she wins and must pay more than she wanted. If she bids six instead, Ren wins, so Mila does not incur a payment.

The rule therefore gives a bid two connected roles. It helps determine whether someone wins, while the other bids determine the price that a winner must accept. To understand a bidder's result, asking only "What did that person offer?" is insufficient. One must also examine the competing bids and the payment rule.

155. In the first example, how many tokens does Mila pay after bidding nine?

A) nine

B) four

C) seven

D) eleven

156. What distinction does the passage chiefly explain?

A) The highest bidder always pays exactly the amount written on that bidder's slip.

B) Every bidder must pay whether or not that bidder wins.

C) The lowest bid determines both the winner and the price.

D) A bid can determine who wins while another bid determines the winner's payment.

157. Ren still bids seven and Tess four. If Mila changes her bid from nine to six, which change follows?

A) Mila changes from paying seven to paying nothing.

B) Mila still wins but now pays six.

C) Tess wins because Mila's bid has decreased.

D) Every bidder now has to pay four.

158. In "Mila does not incur a payment," incur most nearly means ____.

A) become obligated to make

B) secretly increase

C) divide into equal parts

D) receive as a prize

159. How do the two numerical examples work together?

A) The first states that all bidders pay; the second confirms that rule.

B) The first describes a tied auction; the second explains how the tie is broken.

C) The second replaces the original payment rule with a rule charging every bidder's own bid.

D) The first explains why a higher winning bid need not raise the charge; the second shows why overbidding can still be costly.

Literary Passage: An Excellent Pupil

Iris calls me her cat, although anyone observing our apartment carefully would recognize who manages it. I sit beside my bowl; she fills it. A little slow sometimes, but she is learning.

Then she brought home a box with a dish attached. She poured food into its upper compartment and turned a dial to six.

"Dinner at six every evening," she said, "even when we're out."

Humans explain things at unnecessary length. I inspected the box and began its instruction. I tapped the dish with one paw. Nothing. I gave it a longer stare. Nothing. Finally, as the clock struck six, I flicked my tail. Food rattled into the dish.

Excellent. The pupil had recognized the signal.

The following evening, Iris shut me in the hall while she brought in a large parcel. I endured the indignity of waiting beside a pair of boots. The clock struck six beyond the closed door. A moment later, she let me through. The dish was already full.

I considered the evidence carefully. The box could not have seen my tail from the hall. Therefore, it must now understand my thoughts. This was quicker progress than I had expected.

Iris stroked my head. "The timer worked."

I allowed her the remark. A good teacher does not insist on receiving credit for every achievement of an excellent pupil.

160. Where is the narrator when the dish fills on the second evening?

A) beside the food dish

B) outside the apartment building

C) in the hall beside the boots

D) on top of the parcel

161. In "the indignity of waiting beside a pair of boots," indignity most nearly means ____.

A) an accurate measurement

B) a welcome surprise

C) a difficult physical exercise

D) humiliating treatment

162. If the dial stays unchanged and the narrator flicks its tail at five the next evening, what does the passage most strongly support?

A) The machine must immediately dispense dinner at five.

B) Iris must reset every clock in the apartment.

C) The dish will remain empty for the entire evening.

D) The machine will still dispense dinner at six.

163. What does calling Iris "a little slow" and the box "the pupil" reveal about the narrator's point of view?

A) The narrator believes Iris has forbidden all learning.

B) The narrator assumes a superior position as the household's teacher.

C) The narrator regards itself as unable to influence anyone.

D) The narrator describes the apartment from an impartial outside perspective.

164. Why is the narrator's final willingness to let Iris mention the timer ironic?

A) The narrator graciously gives up credit for an event the timer already explains without its teaching.

B) Iris has admitted that the box contains no food.

C) The narrator has successfully changed the dial by thinking.

D) The box has asked Iris to become its teacher.

Scientific Passage: Turning While Going Around

The Moon presents approximately the same face toward Earth as it travels around our planet. Two mistaken conclusions can follow: that the Moon does not rotate, or that its far side must remain without sunlight. Neither conclusion follows from the observation.

To examine rotation, imagine walking around a chair while keeping your face toward it. At each new position, you must turn your body slightly. After one complete circuit, your body has made one complete turn relative to the room. If instead you keep facing one fixed wall throughout the circuit, you will face the chair at some positions and turn your back to it at others.

The Moon similarly rotates once during each trip around Earth, keeping roughly the same side directed toward us. The imagined walker is an explanatory device: it separates turning about one's own center from traveling around another object. The two motions can occur together, with their periods matched.

Illumination is a different question. The far side is called far because it faces away from Earth, not because sunlight never reaches it. As the Moon moves and turns, different portions receive sunlight. Being hidden from an Earthbound observer does not mean being permanently unlit.

This motion model explains how a familiar view can coexist with rotation. It does not explain the history of how the periods became matched; describing an arrangement and explaining its development are separate tasks.

165. In the chair example, what must the walker do while keeping a face toward the chair for a full circuit?

A) remain facing the same wall throughout

B) avoid turning the body at every position

C) make one complete turn relative to the room

D) walk only halfway around the chair

166. What is the passage's main purpose?

A) Prove that the Moon always faces one wall in space

B) Explain the historical process that first matched the Moon's periods

C) Explain why a nearly constant Earth-facing side implies neither absence of rotation nor permanent darkness on the far side

D) Argue that an object can receive sunlight only when Earth can see it

167. A model Moon goes around a model Earth while a painted arrow on the Moon keeps pointing toward one fixed wall. What would an observer at the model Earth see?

A) The same marked side throughout the entire circuit

B) Different sides of the model Moon at different positions

C) No model Moon at any point in the circuit

D) A side that remains permanently beyond the room's light

168. The imagined walker is an "explanatory device." Here, device most nearly means ____.

A) a motor physically attached to the Moon

B) a defect in the observations

C) a method used to make an idea clearer

D) a unit used to measure sunlight

169. Why does the final paragraph distinguish describing the arrangement from explaining its development?

A) To withdraw the earlier claim that the Moon rotates

B) To show that the chair model supplies a complete history of the Moon

C) To suggest that observations cannot support any explanation

D) To specify which question the model answers and which question it leaves open

Humanities Passage: What a Translation Keeps

An invented poem ends, "The last stitch carries the seam." In its context, the image suggests that careful finishing secures earlier work. A translator faces more than a choice between replacing each word and inventing an unrelated message.

One rendering keeps the stitch and seam, preserving the compact image but leaving an unfamiliar reader to interpret it. Another says, "Finish carefully so earlier work holds." That version makes the implied advice explicit but removes the sewing image. Both retain a central claim while producing different reading experiences.

A brief note could explain the image without replacing it. Yet the note adds a pause that the original line may not have required. Its usefulness depends partly on the audience: readers exploring an unfamiliar poem may welcome that help, while readers following a song may prefer an uninterrupted line.

These choices do not make every version equally faithful. A sentence claiming that finishing never matters would reverse the poem's idea. That is a change in the claim, not merely a different balance between image and explanation.

To evaluate the work, a critic should therefore identify what the translation is trying to preserve and ask whether its choices achieve that purpose without reversing the original meaning. Keeping imagery, clarifying an implication, and maintaining a reading rhythm are related aims, but they are not interchangeable tests.

170. What does the first rendering keep that the explanatory version removes?

A) a claim that finishing does not matter

B) the stitch-and-seam image

C) a description of a singer's voice

D) an instruction to stop before the last stitch

171. Which statement best expresses the author's position?

A) A translation is faithful only when every image is replaced with explicit advice.

B) Every possible rendering is equally faithful, including one reversing the original claim.

C) Notes always preserve the exact rhythm of the original reading.

D) Translation choices should be judged by their stated aims while preserving the original's central meaning.

172. An edition is intended for readers unfamiliar with the poem who want BOTH its sewing image and help understanding the implication. Which choice best fits those aims?

A) Replace the image with the opposite advice about finishing.

B) Keep the image and supply a brief explanatory note.

C) Remove the image and omit all explanation.

D) Keep only an unrelated description of the audience.

173. As used in "One rendering keeps the stitch and seam," rendering most nearly means ____.

A) version

B) payment

C) tool

D) measurement

174. What purpose does the example claiming that finishing never matters serve?

A) It distinguishes acceptable differences in emphasis from reversal of the poem's claim.

B) It recommends the most accurate translation of the invented line.

C) It proves that notes are always unnecessary.

D) It shows that the poem is actually a sewing manual.

Section 4: Mathematics

64 questions | Recommended time: 45 minutes

175. Which equality illustrates regrouping in addition without changing the order of the numbers?

A) 3 + 5 = 5 + 3

B) 3 × (5 + 7) = 3 × 5 + 3 × 7

C) 3 + 0 = 3

D) (3 + 5) + 7 = 3 + (5 + 7)

176. Which expression has no real-number value?

A) $-\sqrt{16}$

B) $\sqrt{(-16)}$

C) $\sqrt{(16/4)}$

D) $(-4)^2$

177. Positive real numbers a and b satisfy 2 < a < 3 and 4 < b < 5. Which interval must contain a/b?

A) 1/2 < a/b < 3/5

B) 2/5 < a/b < 3/5

C) 2/5 < a/b < 3/4

D) 1/2 < a/b < 3/4

178. What is the value of $(3^2)^3 \div 3^3$?

A) 9

B) 243

C) 27

D) 729

179. Which choice is an inequality rather than an equation or an expression?

A) 2x + 5 = 17

B) 2x + 5

C) 17 − 5

D) 2x + 5 < 17

180. Which ordered pair (x, y) satisfies BOTH y > 2x and x + y ≤ 7?

A) (2, 5)

B) (3, 4)

C) (2, 6)

D) (0, 0)

181. The equation $x^2 + kx + 12 = 0$ has x = 3 as a solution. What is its other solution?

A) −4

B) 4

C) −3

D) 7

182. A function has domain {−3, −1, 1, 3} and rule f(x) = x². What is its range, the set of distinct outputs?

A) {1, 9}

B) {−3, −1, 1, 3}

C) {−9, −1, 1, 9}

D) {1, 3}

183. Two distinct points are marked in a plane. How many straight lines pass through both?

A) Zero

B) Exactly one

C) Exactly two

D) Infinitely many

184. A circle of radius 3 units lies inside a circle of radius 8 units and touches it at exactly one point. What is the distance between their centers?

A) 5 units

B) 11 units

C) 8 units

D) 3 units

185. A five-tip star is traced in one continuous path, changing direction only at its tips. Each turn has the same size and direction, and the direction completes two full 360° turns altogether. What is the smaller angle between the two segments meeting at a tip?

A) 72°

B) 36°

C) 144°

D) 216°

186. An engineering drawing has scale 1:60. It is printed at 150% of its original linear dimensions. What is the new scale?

A) 1:40

B) 1:90

C) 1:60

D) 1:120

187. A detector uses a scale in which each increase of 2 display units means a fivefold increase in measured intensity. If the display rises from 3 to 9, what is the intensity multiplier?

A) 15

B) 30

C) 25

D) 125

188. Which recorded variable is a numerical measurement rather than an identifying code?

A) a postal code

B) a bus route number

C) an object's mass in grams

D) an athlete's jersey number

189. Reader A decodes 90% of clear codes and 20% of blurred codes correctly; reader B gets 80% and 10%, respectively. A reads 10 clear and 100 blurred codes; B reads 100 clear and 10 blurred codes. Which comparison is correct?

A) A has higher accuracy within each category and overall.

B) B has higher accuracy within each category and overall.

C) A has higher accuracy within each category, but B has higher accuracy overall.

D) The two readers have equal overall accuracy.

190. A fair coin selects jar A or jar B. A contains 3 green tokens and 1 white token; B contains 1 green and 1 white. A token is drawn uniformly from the selected jar and is green. What is the probability that jar A was selected?

A) 1/2

B) 3/4

C) 3/8

D) 3/5

191. On the real number line, how does −0 compare with 0?

A) They name the same point.

B) −0 is one unit to the left.

C) −0 is positive, while 0 is negative.

D) Their order depends on the scale used.

192. Which sum is rational even though both of its displayed addends are irrational?

A) $\sqrt{2} + \sqrt{2}$

B) $\sqrt{2} + (3 - \sqrt{2})$

C) $\sqrt{3} + \sqrt{3}$

D) $\sqrt{5} + (2 + \sqrt{5})$

193. The notation 0.999… means that the digit 9 repeats forever, not that the decimal stops. What is its exact value?

A) a number less than 1 by a final decimal place

B) 1

C) 0.999

D) 9/10

194. For any real numbers a and b, which expression equals the SMALLER of a and b?

A) $(a + b + |a - b|)/2$

B) $(a + b - |a - b|)/2$

C) $(a + b)/2$

D) $|a - b|/2$

195. Which product is equivalent to xy + x + y + 1 for all x and y?

A) $(x + y)(x + 1)$

B) $(x + 1)(y - 1)$

C) $(x + 1)(y + 1)$

D) $xy(x + y + 1)$

196. What real value of x satisfies 2 raised to x = 8 raised to (x − 2)? Equal positive powers of the same base 2 have equal exponents.

A) 1

B) 2

C) 6

D) 3

197. A machine replaces an input x with x/2 + 6. Which input is left unchanged by the machine?

A) 12

B) 3

C) 6

D) 18

198. Two lines in space neither intersect nor lie in the same plane. Such lines are called skew. Which description must be true of skew lines?

A) They meet at a right angle.

B) They are parallel lines in one plane.

C) They are not coplanar.

D) They are the same line.

199. Reflect point (4, −3) across the x-axis and then across the y-axis. Which single transformation has the same effect?

A) a reflection across the line y = x

B) a 180° rotation about the origin

C) a translation 8 units to the right

D) a 90° clockwise rotation about the origin

200. Two concentric circles have radii R and r, with R > r. A chord of the outer circle is tangent to the inner circle and has length 10. What is the area between the circles?

A) 10π square units

B) 50π square units

C) 100π square units

D) 25π square units

201. One degree contains 60 arcminutes. How many degrees are 2 degrees 30 arcminutes?

A) 2.3°

B) 2.05°

C) 2.6°

D) 2.5°

202. A microliter is 0.001 cubic centimeter. A cube-shaped chamber holds exactly 1 microliter. What is the length of each inside edge?

A) 0.001 centimeter

B) 0.01 centimeter

C) 0.1 centimeter

D) 1 centimeter

203. A material requires 200 joules per kilogram for each 1°C temperature increase. With no energy loss or change of state, 8,000 joules heat a 5-kilogram sample. What is its temperature increase?

A) 0.125°C

B) 40°C

C) 8°C

D) 200°C

204. A cumulative tally says "18 measurements are at most 30 seconds." What does this statement mean?

A) Eighteen measurements are 30 seconds or shorter.

B) Exactly eighteen measurements equal 30 seconds.

C) Every measurement is longer than 30 seconds.

D) Eighteen measurements fall between 29 and 30 seconds only.

205. A scatter plot contains the points (−2, 4), (−1, 1), (0, 0), (1, 1), and (2, 4). Which description best fits the relationship?

A) A curved pattern falls to a minimum and then rises.

B) A single increasing straight line passes through all points.

C) A single decreasing straight line passes through all points.

D) All y-values are equal.

206. A coin has probability 4/5 of heads on each independent toss. It is tossed twice, and you are told the two results differ. What is the probability that the FIRST toss was heads?

A) 4/5

B) 1/5

C) 16/25

D) 1/2

207. Which estimate is closest to 1/198 + 1/202?

A) 0.01

B) 0.0001

C) 0.1

D) 1.0

208. A design uses 300 strips, each exactly 2/3 meter long. A planning program substitutes 0.67 meter for every strip. By how much does its computed total exceed the exact total?

A) 0.0033 meter

B) 0.3 meter

C) 1 meter

D) 3 meters

209. Write $(6 \times 10^3) + (7 \times 10^2)$ in scientific notation with a coefficient at least 1 and less than 10.

A) 6.07×10^3

B) 13×10^3

C) 6.7×10^2

D) 6.7×10^3

210. For how many positive integers n are BOTH n and n + 24 perfect squares?

A) 1

B) 2

C) 4

D) Infinitely many

211. Which expression has the same value for EVERY real value of x?

A) x + 4

B) 4x

C) (x + 4) − x

D) $x^2 + 4$

212. Two rules are f(x) = 2x − 5 and g(x) = x^2. What is g(f(4)), found by using f(4) as the input to g?

A) 27

B) 9

C) 11

D) −1

213. A function satisfies f(2) = 5 and f(3) = 7. No other rule is given. What can be concluded about f(5)?

A) Its value cannot be determined from those facts.

B) Its value must be 12.

C) Its value must be 11.

D) Its value must be 17.

214. For which complete set of x-values do the two rules y = x^2 and y = x + 2 give the same output?

A) {1, 2}

B) {−2, 1}

C) {−1, 1}

D) {−1, 2}

215. A convex polyhedron satisfies V − E + F = 2, where V, E, and F count vertices, edges, and faces. If V = 10 and E = 15, how many faces does it have?

A) 5

B) 12

C) 27

D) 7

216. Triangle ABC has A = (0, 0), B = (4, 0), and C = (6, 3). The altitude from C is perpendicular to the LINE through AB. Where does it meet that line?

A) (4, 0), at B

B) (6, 0), beyond B

C) (3, 0), between A and B

D) (0, 3), beside C

217. Points A and B are 10 units apart. Which set consists of ALL points P for which PA + PB = 10?

A) the entire straight line through A and B

B) the segment AB, including A and B

C) the circle with center A and radius 10

D) the perpendicular bisector of segment AB

218. A sealed package is moved from Earth to the Moon without changing its contents. Local gravity is weaker on the Moon. Which quantity remains unchanged?

A) the package's gravitational weight in newtons

B) the local gravitational acceleration

C) the package's mass in kilograms

D) the force needed to support the package at rest

219. On a flat floor, a robot travels 5 meters east, then 12 meters north, then returns directly to its starting point. How much TOTAL distance does its travel recorder register?

A) 0 meters

B) 13 meters

C) 30 meters

D) 17 meters

220. In a survey, each person may choose several preferred fabrics. The percentages choosing wool, cotton, and linen total 170%. Which conclusion is justified?

A) The survey must contain an arithmetic error.

B) Exactly 70% of the people chose all three fabrics.

C) Every person chose at least two fabrics.

D) The total can exceed 100% because one person can be counted in several categories.

221. Eleven integer measurements are grouped as follows: 0–9 has 2 observations; 10–19 has 6; 20–29 has 3. What can be concluded about the median?

A) It is between 10 and 19, inclusive.

B) It is exactly 15.

C) It is exactly 6.

D) It is between 20 and 29, inclusive.

222. A point (x, y) is selected uniformly from the square $0 \le x \le 1$ and $0 \le y \le 1$, so equal areas are equally likely. What is the probability that $y < x/2$?

A) 1/4

B) 1/2

C) 1/8

D) 3/4

223. Which number multiplied by 1 3/5 equals 1?

A) 8/5

B) 3/5

C) 5/8

D) 5/3

224. The decimal 0.416416416… repeats the block 416 forever. What is the sum of the FIRST 20 digits after the decimal point?

A) 71

B) 66

C) 70

D) 77

225. Pieces of lengths 4 and 7 units may be joined end to end, using any nonnegative whole number of each. What is the LARGEST positive integer length that cannot be made?

A) 17

B) 13

C) 18

D) 21

226. Terms are separated by addition or subtraction outside parentheses. How many terms does $3x^2 - 4x + 7$ have?

A) 2

B) 4

C) 3

D) 7

227. Positive numbers p, q, and r satisfy $p = qr^2$. Which formula gives r?

A) p/q^2

B) $(\sqrt{p})/q$

C) p^2/q

D) $\sqrt{(p/q)}$

228. For real inputs, $g(x) = 1/(x^2 - 9)$. Which inputs must be excluded from its domain?

A) 3 only

B) −9 and 9

C) −3 and 3

D) 0 only

229. Which interval gives ALL real solutions of $(x - 2)(x + 3) < 0$?

A) $x < -3$ or $x > 2$

B) $x < -3$

C) $x > 2$

D) $-3 < x < 2$

230. A polygon does not cross itself and has an interior angle of 225°. Which statement must be true?

A) It is concave.

B) It is a regular polygon.

C) Every interior angle is 225°.

D) It has exactly three sides.

231. On a standard analog clock at exactly 3:20, what is the smaller angle between the hour and minute hands?

A) 10°

B) 20°

C) 30°

D) 120°

232. A square has side length 10. At EACH corner, cut along the segment joining points 3 units from that corner on its two adjacent sides. What is the perimeter of the remaining octagon?

A) $40 + 12\sqrt{2}$ units

B) $16 + 6\sqrt{2}$ units

C) $16 + 12\sqrt{2}$ units

D) 28 units

233. Which sum can represent a single physically meaningful measurement after converting units, without changing the type of quantity?

A) 3 meters + 40 centimeters

B) 3 meters + 40 square centimeters

C) 3 kilograms + 40 seconds

D) 3 liters + 40 degrees

234. Water fills a straight trough uniformly along its 2-meter length. Its triangular water cross section has base 6 centimeters and height 3 centimeters. If 1 liter = 1,000 cubic centimeters, how many liters are in the trough?

A) 3.6 liters

B) 0.018 liter

C) 1,800 liters

D) 1.8 liters

235. Counting CALENDAR DAYS, a period runs from the beginning of a month to the beginning of the following month. Neither month nor year is specified. Which statement about the number of days is correct?

A) It must be exactly 30 days.

B) It must be exactly four weeks.

C) It must be exactly 31 days.

D) It can be 28, 29, 30, or 31 days.

236. Which decimal is a possible probability?

A) −0.25

B) 0.65

C) 1.05

D) 2.0

237. A population contains 72 marked animals. A later sample of 30 contains 6 marked animals. A model assumes the marked fraction in the sample equals the marked fraction in the population. What population size does it estimate?

A) 144

B) 360

C) 216

D) 432

238. Two fair six-sided number cubes, labeled 1 through 6, are rolled independently. Given that their PRODUCT is even, what is the probability that their SUM is odd?

A) 1/2

B) 2/3

C) 3/4

D) 1/3

Section 5: Language

60 questions | Recommended time: 30 minutes

239. Which version correctly ends an ordinary declarative sentence with the abbreviation etc.?

A) We packed nails, screws, etc.

B) We packed nails, screws, etc..

C) We packed nails, screws, etc.,

D) We packed nails, screws, etc.;

240. Which revision uses a comma to mark the omitted repetition of contains in the second clause?

A) One drawer, contains spoons; the other forks.

B) One drawer contains spoons; the other, forks.

C) One drawer contains, spoons; the other forks.

D) One drawer contains spoons; the, other forks.

241. Which sentence correctly punctuates a question inside parentheses while keeping the main statement declarative?

A) The package arrived (who sent it)? before daybreak.

B) The package arrived (who sent it.) before daybreak?

C) The package arrived (who sent it?) before daybreak.

D) The package arrived (who sent it) before daybreak?

242. A fictional ruler is called King Alden the Fourth. Which form uses standard capitalization for the Roman numeral in the ruler's name?

A) King Alden IV

B) King Alden iv

C) King Alden Iv

D) King Alden iV

243. Which version uses standard headline-style capitalization for this book title, including its hyphenated modifier?

A) The Self-Guided Journey

B) The Self-guided Journey

C) The self-Guided Journey

D) The self-guided Journey

244. Which spelling correctly completes the sentence?
Please do not ____ the height of the wave.

A) exagerate

B) exaggerrate

C) exaggerate

D) exaggerait

245. Choose the correctly spelled word for concealment achieved by blending with surroundings.

A) camouflage

B) camoflage

C) camoflauge

D) camouflague

246. Which pair correctly completes the sentence?
We need ____ route, but ____ routes remain closed.

A) other; another

B) other; other

C) another; other

D) another; another

247. On Monday, Dora told her brother, "I will deliver the film tomorrow." On Tuesday morning, before the delivery, which report by her brother preserves Dora's meaning?

A) Dora said she would deliver the film tomorrow.

B) Dora said her brother would deliver the film today.

C) Dora said she had delivered the film yesterday.

D) Dora said she would deliver the film today.

248. Which verb phrase correctly completes the sentence?
There ____ two gaps in the fence.

A) is

B) are

C) was

D) has been

249. Which pair correctly completes the sentence?
These are my ____ gloves; I use them ____.

A) every day; everyday

B) everyday; everyday

C) every day; every day

D) everyday; every day

250. An indirect reference to a familiar poem is an ____; a misleading visual appearance is an ____.

A) illusion; allusion

B) allusion; allusion

C) allusion; illusion

D) illusion; illusion

Editing Passage: Help Under a Flap

(1) Why is a hint card folded? Separate flaps let a solver choose how much help to uncover. (2) The first flap points to a specific clue without interpreting it. (3) The next flap explains a relationship among clues but leaves the final step for the solver. (4) The innermost flap contains the complete solution and its reasoning. (5) Someone who needs only a nudge can stop before seeing that solution. (6) What makes a useful first hint? It identifies a relevant feature instead of merely saying, "Try harder." (7) Printing the full solution on the outside would remove the solver's choice before the solver had decided to read it.

251. Which additional FAQ question would be answered by sentences 2–5?

A) What weight of paper should the card use?

B) Who was the first person to solve the puzzle?

C) How many copies of each card should be printed?

D) How much help does each successive level provide?

252. A puzzle hides a city name in the initials of every other line. Which proposed FIRST hint follows the guidance in sentences 2 and 6?

A) Join the initials of the odd-numbered lines in order.

B) The hidden city is Lima, spelled by the selected letters.

C) You need to concentrate and work harder.

D) Notice the letters at the beginnings of the lines.

253. A proposed revision says, "Put the full solution on the outside for easy access." Which replacement preserves access to the solution AND the solver's control over when to see it?

A) Remove the full solution from every version of the card.

B) Keep the solution under the last flap, which the solver may open when ready.

C) Put the full solution outside, but hide the first hint under a flap.

D) Print the hints and full solution together on the outside.

254. An album label shortens the year 1984 to its last two digits. Which version correctly marks the omitted digits?

A) Summer of 84'

B) Summer of 8'4

C) Summer of '84

D) Summer of '84'

255. Which version uses a hyphen to make clear that a sofa will receive a NEW COVER, rather than be retrieved?

A) We will re-cover the sofa.

B) We will recover the sofa.

C) We will re cover the sofa.

D) We will rec-over the sofa.

256. Which form correctly capitalizes the name of this religious text?

A) a translation of the quran

B) a Translation of the quran

C) a translation Of the Quran

D) a translation of the Quran

257. Which sentence uses standard capitalization for an ordinary reference to a century?

A) The design dates from the Twenty-First Century.

B) The design dates from the twenty-first Century.

C) The design dates from the Twenty-first century.

D) The design dates from the twenty-first century.

258. Which spelling correctly completes this introduction?
She is an ____ from my former neighborhood.

A) acquaintance

B) acquantance

C) acquaintence

D) aquaintance

259. Which spelling correctly names the fragrant brown spice used in this filling?

A) cinamon

B) cinnamon

C) cinnammon

D) cinnamin

260. Which spelling names the dark outline of a figure against a lighter background?

A) silouette

B) silhouette

C) silhuoette

D) silhouete

261. Which word correctly introduces the alternatives before this infinitive?
We must decide ____ to wait or leave now.

A) if

B) that

C) unless

D) whether

262. Which sentence correctly forms an exclamation about the size of a single wave?

A) How a huge wave that was!

B) What huge wave that was!

C) What a huge wave that was!

D) How huge a wave that were!

263. Which revision preserves the meaning of "The crate is too wide to fit through the hatch"?

A) The crate is so wide that it cannot fit through the hatch.

B) The crate is so wide that it must fit through the hatch.

C) The crate is wide enough to fit through the hatch.

D) The crate could fit through the hatch if it became wider.

264. Which pair correctly completes the sentence?
The two lamps flashed ____, taking turns. ____, we could leave both switched off.

A) alternately; Alternatively

B) alternatively; Alternately

C) alternately; Alternately

D) alternatively; Alternatively

265. Which pair correctly completes the sentence?
Ari and Bo received the silver and bronze medals, ____. Both thanked the judge ____.

A) respectfully; respectively

B) respectively; respectively

C) respectfully; respectfully

D) respectively; respectfully

Composition Passage: Questions for a Handle Designer

(1) Interviewer: Your first door-handle prototype was smooth and narrow. (2) Designer: Yes. It looked neat, but people wearing thick gloves had trouble gripping it. (3) Interviewer: So you made the next version larger? (4) Designer: Larger, with ridges; those were two changes, not one. (5) Interviewer: Then the test proved that size alone solved the problem. (6) Designer: No. We tested the combined changes, so we cannot separate their effects from that test. (7) Interviewer: What will you try next? (8) Designer: We will compare handles of the same size with and without ridges. That should help us investigate the ridges without changing size at the same time.

266. An editor converts sentences 3–4 into narration. Which version accurately preserves who proposed the idea and who supplied the correction?

A) The designer asked for a larger handle, and the interviewer reported that ridges had been removed.

B) The interviewer asked about a larger handle, and the designer explained that ridges had also been added.

C) Both speakers agreed that size was the only feature changed.

D) The interviewer explained that the first prototype had already included ridges.

267. Why should an edited transcript retain sentence 6 even if sentence 5 is retained as the interviewer's remark?

A) It confirms that the designer agrees with every word of sentence 5.

B) It supplies an unrelated anecdote that interrupts the technical explanation.

C) It prevents a mistaken claim in the interviewer's question from appearing to be the interview's accepted conclusion.

D) It changes the topic from handle design to the interviewer's career.

268. Which follow-up question would challenge a limitation of the PLANNED comparison in sentence 8 without claiming that the comparison has already been performed?

A) Why did the completed test prove that every possible handle size works equally well?

B) After testing ridges at one fixed size, how would you check whether they help at other sizes?

C) Why did you decide never to test the ridges separately?

D) How did the first prototype prove that gloves have no effect on gripping?

269. Which version correctly shows that BOTH short and long combine with term to modify effects?

A) short and long-term effects

B) short-and long-term effects

C) short- and long term effects

D) short- and long-term effects

270. Which version best uses standard punctuation to show that a speaker is abruptly cut off in the middle of a word?

A) "Wait, I haven't fin…"

B) "Wait, I haven't fin—"

C) "Wait, I haven't fin?"

D) "Wait, I haven't fin;"

271. The parenthetical statement is a complete sentence standing on its own. Which version places its period correctly?

A) The gauge fell silent. (Its battery had failed.)

B) The gauge fell silent. (Its battery had failed).

C) The gauge fell silent. (Its battery had failed.).

D) The gauge fell silent. (Its battery had failed,)

272. Which sentence correctly distinguishes a family term used as a name from one following a possessive?

A) I asked mom whether my Dad was home.

B) I asked Mom whether my Dad was home.

C) I asked Mom whether my dad was home.

D) I asked mom whether my dad was home.

273. Use a style that capitalizes a shared geographic term in combined formal names. Which form correctly combines the fictional names Lake Vera and Lake Tano?

A) lakes Vera and Tano

B) Lakes Vera And Tano

C) Lakes Vera and Tano

D) Lakes vera and Tano

274. Which spelling correctly names a short printed guide describing a place or service?

A) brochure

B) broshure

C) brochere

D) brocure

275. Select the correctly spelled name of the creamy condiment.

A) mayonaise

B) mayonnaise

C) mayonnaisse

D) maiyonnaise

276. Which pair correctly completes the active and passive versions?
They made us ____. We were made ____.

A) to wait; wait

B) wait; to wait

C) wait; wait

D) to wait; to wait

277. At 7 p.m., the baker expects to finish at 8:30, before the guests arrive at 9. Which verb phrase uses the FUTURE PERFECT to express completion by 9?
By then, the baker ____ the bread.

A) will have finished

B) will finish

C) has finished

D) had finished

278. Replace the tools with them and the noise with it while keeping standard word order: "She put the tools away but could not put up with the noise."

A) She put away them but could not put up with it.

B) She put them away but could not put it up with.

C) She put them away but could not put up with it.

D) She put away them but could not put up it with.

279. Which word means reduce the tension of a heated disagreement, rather than spread light or another substance?

A) diffuse

B) defuse

C) refuse

D) infuse

280. Which pair correctly completes the sentence?
I ____ the smoke, but I am ____ to complain.

A) loath; loathe

B) loathe; loathe

C) loathe; loath

D) loath; loath

281. Which pair expresses important in history and relating to the past, in that order?

A) historical; historic

B) historic; historic

C) historical; historical

D) historic; historical

Editing Passage: Room for a Cloud

(1) Let your pencil follow a cloud without expecting the cloud to wait. (2) Let one side grow heavier than the other, as though the wind has leaned against it. (3) Let a pale space interrupt the gray; not every part of the sky needs to be filled. (4) You can leave the lower edge unfinished, as though the cloud is still arriving. (5) These are invitations to notice differences, not measurements for producing matching shapes, and a wandering line may record your attention better than a perfect circle. (6) Use exactly the same closed outline for every cloud, with every edge made identical to the last.

282. Which revision of sentence 4 preserves its image and meaning while extending the repeated opening of sentences 1–3?

A) Let every lower edge become a matching straight line.

B) Measure the lower edge carefully until the entire outline is closed.

C) Let the lower edge disappear because the drawing has been abandoned.

D) Let the lower edge stay unfinished, as though the cloud is still arriving.

283. Which replacement for sentence 6 best concludes the draft by reinforcing its invitation to individual observation and connecting with the unfinished-cloud image?

A) Leave a little room for the cloud to become something you had not planned.

B) Copy one approved outline until every cloud has exactly the same shape.

C) Weather reports sometimes list rainfall amounts for several cities.

D) Your drawing is complete only when every uncertain edge has been erased.

284. The main title is Stone and Water; the subtitle is A Study of Walls. Which version places a colon at the correct boundary between them?

A) Stone and Water: A Study of Walls

B) Stone: and Water A Study of Walls

C) Stone and Water A: Study of Walls

D) Stone and Water A Study: of Walls

285. Use the abbreviation U.S. with BOTH of its periods. Which question is punctuated correctly?

A) Was the parcel sent from the U.S?

B) Was the parcel sent from the U.S.?

C) Was the parcel sent from the U.S?.

D) Was the parcel sent from the U.S..?

286. Which form capitalizes the geographic adjective while keeping the common breed noun lowercase?

A) German shepherd

B) german shepherd

C) German Shepherd

D) german Shepherd

287. Which advice-column headline uses standard headline-style capitalization, including its final preposition?

A) Things We Can Agree on

B) Things we Can Agree On

C) Things We Can Agree On

D) Things We Can agree On

288. Which spelling correctly completes the sentence?
The shell had a ____ spiral pattern.

A) peculair

B) peculer

C) peculiar

D) pecculiar

289. Choose the correctly spelled name of the small biting insect.

A) mosqueto

B) mosquito

C) moskito

D) mosquitto

290. Which spelling correctly names a person who starts and organizes a business?

A) entrepreneur

B) entreprenuer

C) entrepenur

D) enterpreneur

291. Which pair correctly completes the sentence?
Departure was delayed ____ fog, and boarding stopped ____ the visibility worsened.

A) because; because of

B) because of; because of

C) because of; because

D) because; because

292. Which word correctly completes the relative clause?
This is the cavern ____ the bats spend the winter.

A) which

B) who

C) whose

D) where

293. Which pair forms a standard unreal PAST conditional about an action that did not happen yesterday?
If Nora ____ the crack yesterday, she ____ the case before shipping it.

A) had seen; would have closed

B) would have seen; had closed

C) has seen; would close

D) had seen; will close

294. Which word means acknowledge a source in a written report?

A) site

B) sight

C) cite

D) seat

295. Which pair correctly names an elected governing group and the advice it receives?
The village ____ sought ____ before deciding.

A) counsel; council

B) council; council

C) counsel; counsel

D) council; counsel

296. Choose the pair distinguishing STAFF ADMINISTRATION from INDIVIDUAL PRIVACY.
A ____ officer handles employee records; workers keep ____ diaries at home.

A) personal; personnel

B) personnel; personal

C) personnel; personnel

D) personal; personal

Editing Passage: Timing an Animated Footstep

(1) To synchronize an animated footstep, editors compare the sound cue with a reference event. (2) All offsets are measured from that moment. (3) Positive offsets put the sound after the reference event; negative offsets put it before. (4) An offset of +3 frames means three frames later, while −3 frames means three frames earlier. (5) The reference event is the first picture frame in which the character's foot touches the surface. (6) Zero offset places the sound at that contact; it does not mean that no sound is used. (7) Changing the sound offset does not require moving the picture frame in which the foot lands.

297. Where should sentence 5 be moved so the reference event is defined before sentence 2 refers to "that moment"?

A) Immediately after sentence 3

B) Immediately after sentence 4

C) Immediately after sentence 6

D) Immediately after sentence 1

298. An added sentence says, "A +3 offset always puts the sound at frame 3 of the whole film." Which replacement clarifies the intended meaning without introducing that false absolute position?

A) A +3 offset requires the contact to occur at frame 3 of every film.

B) A +3 offset means three seconds after the film begins.

C) A +3 offset means that the editor must remove all earlier sound.

D) A +3 offset places the sound three frames after the chosen contact frame, wherever that frame occurs in the film.

Answer Key

Question numbers continue across all five sections.

Verbal Skills

1. C	2. C	3. C	4. D	5. A	6. A	7. C	8. D
9. B	10. A	11. A	12. B	13. B	14. B	15. D	16. D
17. B	18. A	19. C	20. B	21. A	22. B	23. B	24. D
25. D	26. C	27. C	28. A	29. C	30. D	31. A	32. C
33. C	34. D	35. C	36. D	37. B	38. A	39. A	40. A
41. D	42. D	43. C	44. B	45. B	46. C	47. A	48. C
49. D	50. B	51. B	52. A	53. D	54. D	55. D	56. C
57. A	58. C	59. B	60. B				

Quantitative Skills

61. D	62. B	63. C	64. A	65. D	66. C	67. B	68. A
69. B	70. C	71. A	72. B	73. D	74. C	75. A	76. C
77. D	78. A	79. D	80. C	81. A	82. B	83. D	84. C
85. B	86. B	87. D	88. A	89. D	90. B	91. D	92. D
93. B	94. A	95. B	96. C	97. C	98. C	99. A	100. C
101. B	102. A	103. C	104. A	105. D	106. B	107. A	108. C
109. D	110. A	111. B	112. D				

Reading Comprehension

113. D	114. D	115. A	116. C	117. C	118. B	119. A	120. A
121. D	122. C	123. B	124. D	125. B	126. A	127. B	128. C
129. B	130. C	131. B	132. A	133. B	134. A	135. D	136. A
137. B	138. C	139. D	140. C	141. D	142. D	143. A	144. C
145. B	146. D	147. B	148. A	149. C	150. B	151. D	152. A
153. C	154. B	155. C	156. D	157. A	158. A	159. D	160. C
161. D	162. D	163. B	164. A	165. C	166. C	167. B	168. C
169. D	170. B	171. D	172. B	173. A	174. A		

Mathematics

175. D	176. B	177. C	178. C	179. D	180. A	181. B	182. A
183. B	184. A	185. B	186. A	187. D	188. C	189. C	190. D
191. A	192. B	193. B	194. B	195. C	196. D	197. A	198. C
199. B	200. D	201. D	202. C	203. C	204. A	205. A	206. D

207. A	208. C	209. D	210. B	211. C	212. B	213. A	214. D
215. D	216. B	217. B	218. C	219. C	220. D	221. A	222. A
223. C	224. A	225. A	226. C	227. D	228. C	229. D	230. A
231. B	232. C	233. A	234. D	235. D	236. B	237. B	238. B

Language

239. A	240. B	241. C	242. A	243. A	244. C	245. A	246. C
247. D	248. B	249. D	250. C	251. D	252. D	253. B	254. C
255. A	256. D	257. D	258. A	259. B	260. B	261. D	262. C
263. A	264. A	265. D	266. B	267. C	268. B	269. D	270. B
271. A	272. C	273. C	274. A	275. B	276. B	277. A	278. C
279. B	280. C	281. D	282. D	283. A	284. A	285. B	286. A
287. C	288. C	289. B	290. A	291. C	292. D	293. A	294. C
295. D	296. B	297. D	298. D				

Detailed Explanations

Verbal Skills

1. HSPT-V13-001 - Answer C: scattered remains
Debris consists of scattered fragments or remains, especially after something has been broken, damaged, or destroyed.

2. HSPT-V13-002 - Answer C: wearisome
A tedious task is tiresome because it is boring or repetitive; the word does not necessarily imply danger or uncertainty.

3. HSPT-V13-003 - Answer C: simultaneous conflicting feelings
Ambivalence involves opposing feelings toward the same choice, such as wanting its advantages while also resisting its disadvantages.

4. HSPT-V13-004 - Answer D: talkative
Reticent describes someone inclined to hold back speech, whereas talkative describes someone inclined to speak freely and at length.

5. HSPT-V13-005 - Answer A: internal
External locates something on the outside; internal locates it on the inside, directly reversing the stated spatial relationship.

6. HSPT-V13-006 - Answer A: vindictive
A magnanimous response rises above resentment and forgives, while a vindictive response seeks revenge for the offense.

7. HSPT-V13-007 - Answer C: daytime
Nocturnal describes activity during the night, while diurnal describes activity during the day, pairing each adjective with its characteristic active period.

8. HSPT-V13-008 - Answer D: movement
An embargo limits trade, while a quarantine limits movement; each imposed restriction is paired with the activity it directly controls.

9. HSPT-V13-009 - Answer B: a timeline
A dictionary typically arranges entries alphabetically; a timeline arranges events chronologically, pairing an organizing rule with a structure that uses it.

10. HSPT-V13-010 - Answer A: treaty
A treaty is a formal agreement between parties; interview, banter, and dialogue name different forms of verbal exchange.

11. HSPT-V13-011 - Answer A: polish; Polish
Lowercase polish concerns making a surface smooth or shiny; Polish identifies a connection with Poland, so capitalization changes the word's meaning.

12. HSPT-V13-012 - Answer B: hiss, clang
Hiss and clang resemble the noises they name; the alternatives concern sound but do not imitate specific noises.

13. HSPT-V13-013 - Answer B: CAB
The legal moves exchange B with a neighbor but never exchange A with C; A therefore stays before C, ruling out CAB.

14. HSPT-V13-014 - Answer B: B
C meets both conditions for ruling out B; the information does not show that either A or C has a lighter, cheaper alternative.

15. HSPT-V13-015 - Answer D: It substitutes an unrelated personal characteristic for evidence about the claim.
Handwriting quality does not assess the orbital claim; the response rejects the author for an unrelated trait instead of evaluating relevant evidence.

16. HSPT-V13-016 - Answer D: mismatch
A discrepancy is a difference between things expected to agree, so mismatch captures the disagreement between the two accounts.

17. HSPT-V13-017 - Answer B: enjoy fully
To savor something is to appreciate and enjoy it attentively; pausing over the bite emphasizes enjoyment rather than speed or postponement.

18. HSPT-V13-018 - Answer A: a relationship of mutual understanding
Rapport is a harmonious connection marked by mutual understanding; it does not require identical opinions, a formal agreement, or rivalry.

19. HSPT-V13-019 - Answer C: fixed in place
Portable equipment can be carried from place to place; equipment fixed in place has the opposite characteristic.

20. HSPT-V13-020 - Answer B: all at once
Piecemeal means in separate portions or stages; introducing everything at once reverses that manner of proceeding.

21. HSPT-V13-021 - Answer A: far off in time
An imminent event is about to occur; an event far off in time reverses that nearness without making a claim about certainty.

22. HSPT-V13-022 - Answer B: public name
Both are adopted names used publicly in place of another name; the analogy concerns substituted identity labels, not professional equipment or location.

23. HSPT-V13-023 - Answer B: one narrow outlet
The lens converges light toward a focus; the funnel converges liquid toward a narrow outlet, preserving a many-paths-to-one-destination relationship.

24. HSPT-V13-024 - Answer D: imperative
Interrogative identifies the asking function of a question; imperative identifies the directing function of a command, pairing each utterance with its sentence type.

25. HSPT-V13-025 - Answer D: limit
A limit marks an endpoint or boundary; a ledge, spur, or bulge is a projecting feature rather than the boundary itself.

26. HSPT-V13-026 - Answer C: seam; seem
Seam and seem sound alike but have different spellings and meanings; the alternatives share spelling, approximate meaning, or only a rhyme.

27. HSPT-V13-027 - Answer C: left; right
Left and right depend on which way an observer faces; the compass directions remain geographically fixed when the observer turns.

28. HSPT-V13-028 - Answer A: Raising volume always raises brightness.
If volume always raises brightness, its adjustment changes the other setting; that directly conflicts with the stated independent controls.

29. HSPT-V13-029 - Answer C: Having a permit is required for entry, but it does not by itself guarantee entry.
Only states a necessary condition: entrants must have permits. It does not say that a permit is sufficient or that entering is compulsory.

30. HSPT-V13-030 - Answer D: Each claim is offered as support for the other without an independent reason.
The justification circles back to its starting claim; neither statement supplies an independent standard or observation establishing the design's superiority.

31. HSPT-V13-031 - Answer A: required target amount
A production quota specifies an amount expected or required to be produced; it is a target, not a supply, reward, or price.

32. HSPT-V13-032 - Answer C: lighthearted humor
Levity is a lack of solemn seriousness expressed through lightness or humor; it does not necessarily imply hostility or concealed information.

33. HSPT-V13-033 - Answer C: the collection of works the musician can perform
A repertoire consists of the works someone is prepared to perform; it concerns available artistic material rather than equipment, listeners, or maintenance.

34. HSPT-V13-034 - Answer D: imported
Domestic goods are produced within the country under discussion; imported goods come into that country from elsewhere.

35. HSPT-V13-035 - Answer C: clean and well-kept
Squalid conditions are filthy and neglected; clean, well-kept conditions reverse those qualities, whereas size, distance, and liveliness do not.

36. HSPT-V13-036 - Answer D: harmful
An innocuous substance is harmless; harmful reverses that quality without confusing unfamiliarity, appearance, or effectiveness with danger.

37. HSPT-V13-037 - Answer B: the value of sales
Each payment is paired with the amount serving as its calculation base: principal for interest and sales value for the specified commission.

38. HSPT-V13-038 - Answer A: A new organism emerges through its enclosing covering.
Hatching and sprouting involve emergence through an enclosing shell or coat; neither pair describes thickening, retreat, or acquiring a permanent skeleton.

39. HSPT-V13-039 - Answer A: position on a screen
Both serve as position indicators within an information space; neither is the information itself or a measure of speed.

40. HSPT-V13-040 - Answer A: fuel
Fuel is supplied to the engine; the exhaust, ash, and sawdust are described as outputs or residues of their respective processes.

41. HSPT-V13-041 - Answer D: smog; brunch
Smog blends smoke with fog; brunch blends breakfast with lunch. The alternatives join complete words instead of blending shortened parts.

42. HSPT-V13-042 - Answer D: knock; write
The initial k in knock and w in write are silent; the alternatives have silent letters elsewhere or a pronounced initial consonant in one word.

43. HSPT-V13-043 - Answer C: The matching claims share one underlying source.
Both articles transmit the same announcement, so their agreement does not add a separate verification; shared dependence does not itself establish falsity.

44. HSPT-V13-044 - Answer B: Cart R contains no parcels.
An empty cart provides no parcel violating the universal rule; each other situation supplies a parcel on Cart R without a red label.

45. HSPT-V13-045 - Answer B: Rae believes that the blue box is empty.
Ivo's knowledge establishes the fact of Rae's belief, not the truth of its contents; the box may still contain something.

46. HSPT-V13-046 - Answer C: a place of safety
A haven provides shelter or safety; the storm context emphasizes a secure place rather than a warning, journey, or disturbance.

47. HSPT-V13-047 - Answer A: failure
A lapse in attention is a temporary failure to remain attentive; it names the break in performance rather than a demand or improvement.

48. HSPT-V13-048 - Answer C: a sign that change is approaching
A harbinger signals the approach of something; it points forward to a coming development without necessarily causing or preventing it.

49. HSPT-V13-049 - Answer D: rural
Urban concerns a city or town; rural concerns the countryside, whereas coastal location and industry can occur in either setting.

50. HSPT-V13-050 - Answer B: evenly balanced
A lopsided display is unevenly weighted or arranged toward one side; an evenly balanced display reverses that imbalance.

51. HSPT-V13-051 - Answer B: capable of making mistakes
Infallible means incapable of error; its opposite allows mistakes without claiming that every decision must be wrong or that decisions are impossible.

52. HSPT-V13-052 - Answer A: plug
The relationship contrasts a device permitting movement with one blocking it: hinge and latch for a door, open valve and plug for flow.

53. HSPT-V13-053 - Answer D: seller
A monologue has a single speaker despite possibly many listeners; a monopoly has a single seller despite possibly many buyers.

54. HSPT-V13-054 - Answer D: boat
A foot leaves a footprint as it passes; a boat leaves a wake in the water, pairing a residual passage trace with its source.

55. HSPT-V13-055 - Answer D: stride
Stride names a step or the act of striding; tremor, shudder, and quiver name shaking or trembling movements.

56. HSPT-V13-056 - Answer C: glare, glow, gleam
Glare, glow, and gleam describe forms of light or shine; the other groups name colors, coloring materials, or the absence of light.

57. HSPT-V13-057 - Answer A: unhyphenated
Unhyphenated contains no hyphen; uppercase is printed in lowercase, hyphenated has no hyphen, and THREE WORDS contains only two words.

58. HSPT-V13-058 - Answer C: It excludes a selective option that is available.
The reviewer treats total acceptance and total refusal as exhaustive, but the stated selective approach supplies a third available course.

59. HSPT-V13-059 - Answer B: The two segments may meet at an angle.
Straightness of the individual segments does not fix their directions relative to one another; an angle between them makes the combined route bend.

60. HSPT-V13-060 - Answer B: Bea's badge is red.
If Bea wore blue, Ari would see it and know that Ari must wear red. Ari's uncertainty therefore tells Bea that Bea wears red.

Quantitative Skills

61. HSPT-Q13-001 - Answer D: 840
The next term must include 8 as well as all factors already represented by 420. Their least common multiple is 840.

62. HSPT-Q13-002 - Answer B: 6
The last 1 is at position 9; its most recent earlier occurrence is at position 3. Their distance is 9 − 3 = 6.

63. HSPT-Q13-003 - Answer C: 17/41
Adding 2 to 7/17 gives 41/17; taking its reciprocal gives 17/41, continuing the same two-stage fraction transformation.

64. HSPT-Q13-004 - Answer A: 6174
For 8352, the descending arrangement is 8532 and the ascending arrangement is 2358; subtracting gives 8532 − 2358 = 6174.

65. HSPT-Q13-005 - Answer D: Every positive integer
For every positive integer d, choosing k = 0 gives 0 = d × 0, so every such d meets the definition.

66. HSPT-Q13-006 - Answer C: 9 samples against 9 samples
Weighing k against k leaves candidate groups of sizes k, k, and 27 − 2k. At k = 9, all three are 9, minimizing the largest group.

67. HSPT-Q13-007 - Answer B: 17
Nine have tens digit 5 but are not 55. Eight have ones digit 5 with a different nonzero tens digit, giving 9 + 8 = 17.

68. HSPT-Q13-008 - Answer A: Quantity A is greater.
If every bin held at most one packet, only five packets could be placed. Six packets therefore force some bin to contain at least two.

69. HSPT-Q13-009 - Answer B: Quantity B is greater.
Writing a < b < c < d gives median (b + c)/2 and Quantity B (c + d)/2; their difference is (d − b)/2 > 0.

70. HSPT-Q13-010 - Answer C: The two quantities are equal.
Both the maximum and minimum increase by 7; subtracting them cancels the added amount, so the range stays unchanged.

71. HSPT-Q13-011 - Answer A: Quantity A is greater.
An inscribed angle measures half the central angle intercepting the same arc; the shared arc is nonzero, making Quantity A strictly greater.

72. HSPT-Q13-012 - Answer B: Quantity B is greater.
The angle-bisector rule gives BD/DC = AB/AC = 5/8. Since this ratio is less than 1, BD is shorter than DC.

73. HSPT-Q13-013 - Answer D: The relationship cannot be determined from the information given.
Radius 1, height 5 gives surface area 12π and volume 5π; radius 2, height 1 gives the same area but volume 4π. Either cylinder can be first.

74. HSPT-Q13-014 - Answer C: 1
The next representation is 1000, which contains one digit 1. The entries count those digits rather than the representation's total length or value.

75. HSPT-Q13-015 - Answer A: 3/2
The current group has numerator plus denominator equal to 5. After 2/3, increasing the numerator to 3 makes the denominator 2.

76. HSPT-Q13-016 - Answer C: 222
After both one-digit and all four two-digit possibilities, the least three-digit possibility is 222; 227 comes after it.

77. HSPT-Q13-017 - Answer D: It divides by zero.
Because a = b, the canceled factor a − b equals zero. Division by that factor is undefined even though the preceding equality is true.

78. HSPT-Q13-018 - Answer A: Exchange the first and second digits.
The original weighted sum is 40. Exchanging the first two digits gives 7254, with sum 30. The other exchanges give sums 36, 36, and 42, none divisible by 10.

79. HSPT-Q13-019 - Answer D: 6
If the smallest were 7, the least possible sum would be 7 + 8 + 9 + 10 + 11 = 45. The values 6 through 10 total 40.

80. HSPT-Q13-020 - Answer C: 4 and 11
Write the integer as 7q + 4. An even q leaves remainder 4 modulo 14; an odd q leaves remainder 11.

81. HSPT-Q13-021 - Answer A: Quantity A is greater.
The floor is at most x and cannot equal it because x is not an integer; therefore the floor is strictly smaller.

82. HSPT-Q13-022 - Answer B: Quantity B is greater.
Squaring a + b = 1 gives $a^2 + b^2 + 2ab = 1$. Since ab is positive, $a^2 + b^2$ must be less than 1.

83. HSPT-Q13-023 - Answer D: The relationship cannot be determined from the information given.
If the events coincide, their intersection has probability 1/2; if they are complementary, it has probability 0. Both satisfy the stated individual probabilities.

84. HSPT-Q13-024 - Answer C: The two quantities are equal.
A shift preserves area, and either enlargement multiplies area by 4. The operation order can affect position but not either resulting area.

85. HSPT-Q13-025 - Answer B: Quantity B is greater.
Each center-to-vertex segment has length s. Opposite vertices are therefore 2s apart, while three sides total 3s, which is greater.

86. HSPT-Q13-026 - Answer B: Quantity B is greater.
Equal perimeters give equal side lengths s. The rhombus's perpendicular height is less than s, so its area is less than the square's s^2.

87. HSPT-Q13-027 - Answer D: 312211
The last term has three 1s, two 2s, and one 1 in successive runs. Writing those counts and digits gives 31, 22, 11: 312211.

88. HSPT-Q13-028 - Answer A: 13
Candidates 9–12 repeat existing sums: 10, 12, 12, and 16, respectively. Adding 13 creates only new pair sums, so 13 is the next term.

89. HSPT-Q13-029 - Answer D: 4
For denominator 10, the eligible numerators are 1, 3, 7, and 9; each shares no factor greater than 1 with 10.

90. HSPT-Q13-030 - Answer B: 0
The next index is 7. Since $7^2 = 49$ is divisible by 7, its remainder is 0; a remainder cannot equal the divisor.

91. HSPT-Q13-031 - Answer D: 1
Dividing k by itself gives 1; the equal-base exponent rule represents that same quotient as k raised to $1 - 1$, or power 0.

92. HSPT-Q13-032 - Answer D: 1 1/8
In eighths, the lengths are 18 and 27. Their greatest common factor is 9, so the greatest shared unit is 9/8 = 1 1/8.

93. HSPT-Q13-033 - Answer B: 64
If two entries differ by at least 2, moving 1 from the larger to the smaller increases their product. Thus the maximum uses 4, 4, 4: product 64.

94. HSPT-Q13-034 - Answer A: Quantity A is greater.
There are six ordered lists: AB, BA, AC, CA, BC, CB. They form only three unordered groups, so Quantity A is greater.

95. HSPT-Q13-035 - Answer B: Quantity B is greater.
Because 2/3 < 3/4 and both are positive, multiplying n copies preserves the strict ordering for every positive integer n.

96. HSPT-Q13-036 - Answer C: The two quantities are equal.
Substitute b = 1/a. Then 1/(1 + b) = a/(1 + a), so the two fractions add to (1 + a)/(1 + a) = 1.

97. HSPT-Q13-037 - Answer C: The two quantities are equal.
With radius r, Quantity A is $\pi r^2/(2r) = \pi r/2$. Quantity B is $(2\pi r)/4 = \pi r/2$, so they are equal.

98. HSPT-Q13-038 - Answer C: The two quantities are equal.
M is 5 units from either hypotenuse endpoint. The stated equidistance property makes its distance to the third vertex 5 units as well.

99. HSPT-Q13-039 - Answer A: Quantity A is greater.
Equal volumes give $(1/3)\pi R^2 h = \pi r^2 h$, so $R^2 = 3r^2$. Positive radii therefore require R > r, making the cone's radius greater.

100. HSPT-Q13-040 - Answer C: 1
The digits begin 98 | 99 | 100 | 101. The list stops after the first two digits of 101, so the next digit is 1.

101. HSPT-Q13-041 - Answer B: (1, 2, 1)
From (2, 1, 3), the differences are $|2 - 1| = 1$, $|1 - 3| = 2$, and $|3 - 2| = 1$.

102. HSPT-Q13-042 - Answer A: 3
For 15, the sums are 7 + 8, 4 + 5 + 6, and 1 + 2 + 3 + 4 + 5. Longer positive runs already exceed 15.

103. HSPT-Q13-043 - Answer C: 4:1
Two meters equal 200 centimeters. The ratio becomes 200:50, which reduces to 4:1; dividing the unconverted numbers would compare unlike units.

104. HSPT-Q13-044 - Answer A: 28
The proper positive divisors of 28 are 1, 2, 4, 7, and 14; their sum is 28, meeting the stated definition.

105. HSPT-Q13-045 - Answer D: 12
Multiplying the roots gives $\sqrt{(6 \times 24)} = \sqrt{144} = 12$; taking the product inside one root avoids decimal approximations.

106. HSPT-Q13-046 - Answer B: Put 5 with the object and 2 and 11 on the opposite pan.
With 5 beside the object, that pan totals 8 + 5 = 13; the opposite pan totals 2 + 11 = 13.

107. HSPT-Q13-047 - Answer A: Quantity A is greater.
Both numbers are negative, and −0.003 is closer to zero than −0.03; therefore −0.003 is the greater real number.

108. HSPT-Q13-048 - Answer C: The two quantities are equal.
Because the events are disjoint, no outcome is counted twice; their union probability is exactly the sum of the two probabilities.

109. HSPT-Q13-049 - Answer D: The relationship cannot be determined from the information given.
For n = 10, its square 100 has more zeros. For n = 106, its square 11236 has fewer zeros. Thus either ordering is possible.

110. HSPT-Q13-050 - Answer A: Quantity A is greater.
A radius to a point of tangency is perpendicular to the tangent line, so Quantity A is 90°, which exceeds 75°.

111. HSPT-Q13-051 - Answer B: Quantity B is greater.
Half the sum of two different lengths lies strictly between them, so the midsegment is shorter than the longer base.

112. HSPT-Q13-052 - Answer D: The relationship cannot be determined from the information given.
Both triangles share base AB. The farther intersection has greater perpendicular height, but either C or D may be the farther point, so the area ordering is unspecified.

Reading Comprehension

113. HSPT-R13-001 - Answer D: It has been set to take a photograph after ten seconds.
The aunt sets a ten-second delay before joining the row, so the camera clicks automatically during the unplanned reaction.

114. HSPT-R13-002 - Answer D: crookedly
Askew describes the hat's tilted position, contrasting with the careful straightening and upright arrangement that the narrator initially wanted.

115. HSPT-R13-003 - Answer A: It fails the original plan that everyone should look at the camera together.
The narrator judges success by the initial posing instructions; the last image includes everyone but catches their attention turned elsewhere.

116. HSPT-R13-004 - Answer C: The family's stiff, carefully arranged appearance before the interruption
The image of upright books highlights alignment and stillness, preparing a contrast with the spontaneous movement and laughter that follow.

117. HSPT-R13-005 - Answer C: He values the relationships and shared delight revealed by the photograph.
Every image contains the same people; Grandfather's choice values their recognizable interaction rather than merely their physical inclusion in the frame.

118. HSPT-R13-006 - Answer B: It follows a longer path, while each revolution covers the same distance for either wheel.
The passage combines a longer outside path with equal circumferences, so the outside wheel must rotate more times in the shared interval.

119. HSPT-R13-007 - Answer A: How unequal paths in a turn create a need for different rotation rates in equal-sized wheels
The explanation follows the outside and inside paths, connects them to rotations, and identifies the mechanism that permits the needed difference.

120. HSPT-R13-008 - Answer A: Its larger circumference could cover more distance in each revolution.
The final paragraph identifies equal circumference as essential; a larger wheel gains distance per revolution, so path length alone cannot determine the rotation count.

121. HSPT-R13-009 - Answer D: travel along
The outside wheel traverses a longer path by moving along it; the word describes travel rather than dividing or eliminating the route.

122. HSPT-R13-010 - Answer C: It identifies assumptions that limit when revolution counts can directly compare distances.
The final paragraph changes wheel size as a test case, showing why the earlier conclusion depends on equal circumferences and a shared time interval.

123. HSPT-R13-011 - Answer B: two shorter lengths
The first paragraph explicitly contrasts two usable lengths of twine with a hammock that cannot become two working hammocks by the same procedure.

124. HSPT-R13-012 - Answer D: It can test a principle by simplifying a case while leaving practical questions unresolved.
The passage uses the invented case to reveal a distinction, then explains why omitted circumstances limit the practical conclusions it can establish.

125. HSPT-R13-013 - Answer B: Equal cutting might no longer provide either person with the needed benefit.
If both need longer pieces, the halves would fail their purpose, reinforcing the need to consider uses rather than material division alone.

126. HSPT-R13-014 - Answer A: consider separately
Removing extra details lets the reader examine one distinction apart from other complications; isolate describes analytical separation, not destruction or absolute proof.

127. HSPT-R13-015 - Answer B: It frames the example as a way to investigate a rule rather than judge people whose circumstances are unknown.
The concluding contrast preserves the example's investigative value while limiting its authority over real situations that the passage has not described.

128. HSPT-R13-016 - Answer C: who already owns the objects
The third paragraph specifically says the case provides no ownership information, even though the opening states the objects and the proposed division.

129. HSPT-R13-017 - Answer B: whether she can become an apprentice
The opening paragraph identifies the workshop's promise to respond to her apprenticeship request; the later household details are distractions from that answer.

130. HSPT-R13-018 - Answer C: Neri begins to face an uncertain answer after delaying it through small tasks.
The tasks postpone opening the envelope, but the final action begins the reading; the story never reveals whether the workshop accepts her.

131. HSPT-R13-019 - Answer B: He offers her control over his presence instead of pressuring or teasing her.
Neri expects laughter, but her brother asks whether she wants privacy; his respectful choice removes the confrontation she had anticipated.

132. HSPT-R13-020 - Answer A: The tension between the answer's availability and Neri's delay in learning it
The answer has arrived physically, yet Neri postpones knowing it; the recurring delay holds those two conditions apart.

133. HSPT-R13-021 - Answer B: It gives Neri a sense of continuity while the letter's possible answer remains uncertain.
The surroundings remain familiar even as she faces new information; that physical continuity steadies the moment without disclosing the answer.

134. HSPT-R13-022 - Answer A: The attraction between sheets is weaker than the bonds within a sheet.
The second paragraph distinguishes strong connections within sheets from weaker attraction between them, which permits sliding and small pieces rubbing off.

135. HSPT-R13-023 - Answer D: Atomic arrangement can help explain different properties in materials made from the same element.
The passage holds the element constant and compares layered and three-dimensional arrangements to explain contrasting behavior.

136. HSPT-R13-024 - Answer A: Rubbing off can involve separation between sheets without requiring all bonds within each sheet to be weak.
The passage specifically separates strong in-sheet bonding from weaker between-sheet attraction; the visible mark does not establish uniform weakness throughout the structure.

137. HSPT-R13-025 - Answer B: withstands
Diamond resists scratching by withstanding it or being difficult to scratch; the word does not describe measuring or reproducing a scratch.

138. HSPT-R13-026 - Answer C: Explain graphite's layers, explain diamond's network, then draw a broader conclusion about material descriptions
The middle paragraphs connect each structure to behavior; the final paragraph broadens the comparison into a principle about composition and arrangement.

139. HSPT-R13-027 - Answer D: the same melody
The opening defines a round as groups singing the same melody with different starting times, separating shared musical material from staggered entrances.

140. HSPT-R13-028 - Answer C: Delayed entrances let one melody form overlapping musical parts.
The passage explains how the same tune, begun at different times, produces simultaneous portions and a combined texture.

141. HSPT-R13-029 - Answer D: The required overlap of different portions has not been shown.
The passage contrasts sequential echoing with simultaneous portions in a round; repeating material does not by itself establish that overlap.

142. HSPT-R13-030 - Answer D: the way the simultaneous musical parts combine
The surrounding sentences describe overlapping portions sounding together, so texture concerns the combination of musical parts rather than a physical surface or duration.

143. HSPT-R13-031 - Answer A: what the groups are singing simultaneously
The final paragraph explicitly assigns the vertical view to simultaneous sound, while following a horizontal row traces one group through time.

144. HSPT-R13-032 - Answer C: It represents the difference between one part over time and several parts at one time.
Rows show individual melodic progress, while vertical alignment shows combined sound; the visual model clarifies the passage's central relationship.

145. HSPT-R13-033 - Answer B: a ship above three waves
The clerk presses the stamp into wax, and the next sentence explicitly describes its impression as a ship above three waves.

146. HSPT-R13-034 - Answer D: An effort to anticipate a task's needs becomes embarrassing because he misunderstands a key word.
Beto prepares water and food for an animal, then discovers that seal means a stamp; his helpful planning rests on that misunderstanding.

147. HSPT-R13-035 - Answer B: He uses the phrase to avoid explaining the embarrassing reason for his purchase.
Beto is relieved that the stamp is hidden but uneasy about the visible fish, supporting an attempt to conceal his mistake rather than official secrecy.

148. HSPT-R13-036 - Answer A: a mark left by pressing
The stamp is pressed into wax and leaves a visible ship-and-waves design, so impression names the resulting physical mark.

149. HSPT-R13-037 - Answer C: She playfully treats the stamp as the animal Beto expected.
The remark gives an inanimate stamp an animal's appetite, gently exposing the double meaning behind Beto's preparations.

150. HSPT-R13-038 - Answer B: The narrator cannot carry the heavy delivery tricycle down the stairs.
The narrator explicitly connects the steep stairs with the tricycle's heavy front box and the inability to carry it downstairs.

151. HSPT-R13-039 - Answer D: The narrator overcomes reluctance to object and brings an overlooked practical difficulty into the discussion.
The narrator first joins apparent agreement, then speaks about an access problem; the ending opens discussion without announcing a final storage decision.

152. HSPT-R13-040 - Answer A: The discussion reveals that apparent agreement had concealed an access difficulty affecting more than one person.
The narrator's explanation prompts another tenant to identify a similar concern, so the earlier summary no longer represents a fully examined agreement.

153. HSPT-R13-041 - Answer C: It shifts from joining a vote to requesting an opportunity to speak.
The same physical action first follows the group's agreement, then initiates a question, and finally expresses a willingness to keep contributing.

154. HSPT-R13-042 - Answer B: It emphasizes the opening of a more informed conversation rather than claim that the practical problem is already solved.
The final paragraph explicitly leaves both the crowded hall and the stairs unresolved while showing the narrator and another tenant ready to contribute.

155. HSPT-R13-043 - Answer C: seven
Mila has the highest bid, but the rule charges the second-highest bid, which is Ren's seven tokens.

156. HSPT-R13-044 - Answer D: A bid can determine who wins while another bid determines the winner's payment.
Both examples separate ranking from charging and show why understanding the outcome requires the competing bids as well as the stated rule.

157. HSPT-R13-045 - Answer A: Mila changes from paying seven to paying nothing.
At six, Mila falls below Ren and loses; because only the winner pays, her own payment becomes zero rather than her newly offered six.

158. HSPT-R13-046 - Answer A: become obligated to make
As a losing bidder, Mila owes nothing; incur a payment means become responsible for making it, not receive tokens or change the bid.

159. HSPT-R13-047 - Answer D: The first explains why a higher winning bid need not raise the charge; the second shows why overbidding can still be costly.
The examples first separate an offer from its charge, then prevent readers from mistaking that separation for a guarantee that any high offer is safe.

160. HSPT-R13-048 - Answer C: in the hall beside the boots
Iris has shut the narrator in the hall while bringing in a parcel; the clock strikes before she lets the narrator return to the full dish.

161. HSPT-R13-049 - Answer D: humiliating treatment
The self-important narrator regards being shut away beside boots as beneath its dignity, so indignity expresses perceived humiliating treatment.

162. HSPT-R13-050 - Answer D: The machine will still dispense dinner at six.
Iris specifies a daily six-o'clock schedule, and feeding occurs even while the narrator is behind a closed door; the tail gesture is not the stated trigger.

163. HSPT-R13-051 - Answer B: The narrator assumes a superior position as the household's teacher.
The first-person voice assigns the narrator the teacher's role and treats Iris and the machine as learners, establishing its comical self-importance.

164. HSPT-R13-052 - Answer A: The narrator graciously gives up credit for an event the timer already explains without its teaching.
The text supplies the timer as the working explanation, while the narrator treats Iris's accurate remark as modestly tolerable ignorance of its own supposed achievement.

165. HSPT-R13-053 - Answer C: make one complete turn relative to the room
The second paragraph states that continuously facing the chair requires gradual turning, totaling one complete turn relative to the room after a circuit.

166. HSPT-R13-054 - Answer C: Explain why a nearly constant Earth-facing side implies neither absence of rotation nor permanent darkness on the far side
The passage uses the walking model to explain rotation, then separates Earth-facing position from illumination and explicitly limits its historical claims.

167. HSPT-R13-055 - Answer B: Different sides of the model Moon at different positions
The fixed-wall walker faces toward and away from the chair at different positions; the model arrow imposes that same unchanged room-relative orientation.

168. HSPT-R13-056 - Answer C: a method used to make an idea clearer
The walker is an imagined example that clarifies two motions, so device refers to an explanatory method rather than a physical machine or measuring unit.

169. HSPT-R13-057 - Answer D: To specify which question the model answers and which question it leaves open
The ending keeps the model's account of the present motion while expressly separating it from an explanation of how that motion developed.

170. HSPT-R13-058 - Answer B: the stitch-and-seam image
The first version retains stitch and seam, while the second states the advice directly without preserving that sewing image.

171. HSPT-R13-059 - Answer D: Translation choices should be judged by their stated aims while preserving the original's central meaning.
The passage accepts differing priorities among image, explanation, and rhythm but rejects reversing the core claim, linking evaluation to both purpose and fidelity.

172. HSPT-R13-060 - Answer B: Keep the image and supply a brief explanatory note.
The passage identifies a note as a way to explain without replacing the image and says exploratory readers may welcome that help.

173. HSPT-R13-061 - Answer A: version
The surrounding discussion compares ways to express the poem in translation, so a rendering is a particular version rather than a payment or instrument.

174. HSPT-R13-062 - Answer A: It distinguishes acceptable differences in emphasis from reversal of the poem's claim.
After discussing tradeoffs among features, the author sets a limit: preserving different effects does not justify stating the opposite central idea.

Mathematics

175. HSPT-M13-001 - Answer D: (3 + 5) + 7 = 3 + (5 + 7)
Only the parentheses move in the correct equality; the numbers keep their order, illustrating addition's associative property rather than reordering, distribution, or identity.

176. HSPT-M13-002 - Answer B: √(−16)
No real number has a negative square, so √(−16) is not real; the other expressions evaluate to −4, 2, and 16.

177. HSPT-M13-003 - Answer C: 2/5 < a/b < 3/4
The smallest limiting quotient uses the lower numerator and upper denominator, 2/5; the largest uses the upper numerator and lower denominator, 3/4.

178. HSPT-M13-004 - Answer C: 27
Cubing 3^2 gives 3 raised to 6; dividing by 3^3 leaves $3^3 = 27$, so exponents multiply before they subtract.

179. HSPT-M13-005 - Answer D: 2x + 5 < 17
The less-than sign compares unequal quantities, making the correct choice an inequality; the others are an equation or expressions without a comparison.

180. HSPT-M13-006 - Answer A: (2, 5)
For (2, 5), 5 > 4 and 2 + 5 = 7. The alternatives fail either the first strict inequality or the second bound.

181. HSPT-M13-007 - Answer B: 4
Substituting 3 gives 21 + 3k = 0, so k = −7. Then $x^2 - 7x + 12 = (x - 3)(x - 4)$, giving the other solution 4.

182. HSPT-M13-008 - Answer A: {1, 9}
Squaring the four inputs gives 9, 1, 1, and 9; the distinct outputs are therefore just 1 and 9.

183. HSPT-M13-009 - Answer B: Exactly one
Two distinct points determine one straight line; many lines can pass through either point separately, but only one passes through both.

184. HSPT-M13-010 - Answer A: 5 units
At internal tangency, the centers and contact point lie on one line; the center distance is the difference of the radii, 8 − 3 = 5.

185. HSPT-M13-011 - Answer B: 36°
The five equal turns total 720°, so each turn is 144°. The smaller angle between the meeting segments is its supplement, 180° − 144° = 36°.

186. HSPT-M13-012 - Answer A: 1:40
Each printed length becomes 1.5 times as large while the real object stays unchanged; the scale becomes 1.5:60, equivalent to 1:40.

187. HSPT-M13-013 - Answer D: 125
The display increases by 6 units, or three 2-unit steps. Applying the fivefold change three times gives 5 × 5 × 5 = 125.

188. HSPT-M13-014 - Answer C: an object's mass in grams
Mass in grams measures an amount; the other numbers label locations, routes, or people and do not measure their quantities.

189. HSPT-M13-015 - Answer C: A has higher accuracy within each category, but B has higher accuracy overall.
A gets 9 + 20 = 29 of 110 correct; B gets 80 + 1 = 81 of 110. Different category mixes reverse the overall comparison.

190. HSPT-M13-016 - Answer D: 3/5
The A-and-green probability is 3/8; the total green probability is 3/8 + 1/4 = 5/8. Conditioning gives (3/8)/(5/8) = 3/5.

191. HSPT-M13-017 - Answer A: They name the same point.
Zero is its own additive opposite, so −0 equals 0 and names exactly the same point on the real number line.

192. HSPT-M13-018 - Answer B: √2 + (3 − √2)
The opposite square-root terms cancel, leaving 3. The other sums retain nonzero multiples of irrational square roots.

193. HSPT-M13-019 - Answer B: 1
Since 1/3 = 0.333…, multiplying both sides by 3 gives 1 = 0.999…; an endless decimal has no final place containing a gap.

194. HSPT-M13-020 - Answer B: (a + b − |a − b|)/2
If a ≤ b, then |a − b| = b − a and the expression becomes a; reversing their order makes it become b.

195. HSPT-M13-021 - Answer C: (x + 1)(y + 1)
Grouping gives x(y + 1) + (y + 1); taking out the shared factor y + 1 yields (x + 1)(y + 1).

196. HSPT-M13-022 - Answer D: 3
Since $8 = 2^3$, the equation becomes 2 raised to x = 2 raised to (3x − 6). Thus x = 3x − 6, giving x = 3.

197. HSPT-M13-023 - Answer A: 12
An unchanged input satisfies x = x/2 + 6. Subtracting x/2 gives x/2 = 6, so the fixed input is 12.

198. HSPT-M13-024 - Answer C: They are not coplanar.
Coplanar means lying in one plane; the definition explicitly excludes that condition for skew lines.

199. HSPT-M13-025 - Answer B: a 180° rotation about the origin
The reflections produce (4, 3) and then (−4, 3). A half-turn about the origin also changes both coordinate signs.

200. HSPT-M13-026 - Answer D: 25π square units
Half the chord is 5, and its distance from the common center is r. The right triangle gives $R^2 − r^2 = 25$, so the annular area is 25π.

201. HSPT-M13-027 - Answer D: 2.5°
Thirty arcminutes equal 30/60 = 0.5 degree, so the total is 2 + 0.5 = 2.5 degrees.

202. HSPT-M13-028 - Answer C: 0.1 centimeter
A cube's volume is its edge length cubed. Since 0.1 × 0.1 × 0.1 = 0.001, each edge measures 0.1 centimeter.

203. HSPT-M13-029 - Answer C: 8°C
Heating all 5 kilograms by 1°C requires 5 × 200 = 1,000 joules. The supplied 8,000 joules therefore raises the temperature by 8°C.

204. HSPT-M13-030 - Answer A: Eighteen measurements are 30 seconds or shorter.
A cumulative count includes every observation at or below the stated boundary; it does not refer only to values equal or immediately adjacent to it.

205. HSPT-M13-031 - Answer A: A curved pattern falls to a minimum and then rises.
From left to right, y falls from 4 to 0 and rises back to 4, forming a U-shaped pattern rather than a straight trend.

206. HSPT-M13-032 - Answer D: 1/2
Heads-then-tails and tails-then-heads each have probability (4/5)(1/5) = 4/25. Conditioning on these equally likely alternatives gives 1/2.

207. HSPT-M13-033 - Answer A: 0.01
Both fractions are close to 1/200, so their sum is close to 2/200 = 1/100 = 0.01.

208. HSPT-M13-034 - Answer C: 1 meter
The exact total is 300 × 2/3 = 200 meters; the program obtains 300 × 0.67 = 201 meters, an excess of 1 meter.

209. HSPT-M13-035 - Answer D: 6.7×10^3
The terms are 6,000 and 700, totaling 6,700. Keeping the common scale of 1,000 gives the normalized coefficient 6.7.

210. HSPT-M13-036 - Answer B: 2
Write $n = a^2$ and $n + 24 = b^2$. Then (b − a)(b + a) = 24; even factor pairs (2, 12) and (4, 6) give n = 25 and 1.

211. HSPT-M13-037 - Answer C: (x + 4) − x
The two x terms cancel in (x + 4) − x, leaving the constant 4; the other expressions vary as x changes.

212. HSPT-M13-038 - Answer B: 9
First f(4) = 8 − 5 = 3; then $g(3) = 3^2 = 9$, so the first output becomes the second input.

213. HSPT-M13-039 - Answer A: Its value cannot be determined from those facts.
The two facts fix outputs at inputs 2 and 3 only; additivity or a linear rule cannot be assumed for the different input 5.

214. HSPT-M13-040 - Answer D: {−1, 2}
Set $x^2 = x + 2$. Then $x^2 − x − 2 = (x − 2)(x + 1) = 0$, giving x = 2 or x = −1.

215. HSPT-M13-041 - Answer D: 7
Substitute the counts: 10 − 15 + F = 2. Thus F − 5 = 2, so the polyhedron has 7 faces.

216. HSPT-M13-042 - Answer B: (6, 0), beyond B
AB lies on the horizontal line y = 0, so the perpendicular from C keeps x = 6 and meets the extended line at (6, 0).

217. HSPT-M13-043 - Answer B: the segment AB, including A and B
On segment AB, the two distances add to AB. Beyond either endpoint or off the line, their sum is greater than 10.

218. HSPT-M13-044 - Answer C: the package's mass in kilograms
Mass measures the amount of matter and remains unchanged; gravitational weight and the supporting force depend on the weaker local gravity.

219. HSPT-M13-045 - Answer C: 30 meters
The direct return is $\sqrt{(5^2 + 12^2)} = 13$ meters. The total path length is 5 + 12 + 13 = 30 meters.

220. HSPT-M13-046 - Answer D: The total can exceed 100% because one person can be counted in several categories.
Multiple selections let the categories overlap, so their percentages need not sum to 100%; the total alone does not determine individual combinations.

221. HSPT-M13-047 - Answer A: It is between 10 and 19, inclusive.
The median is the sixth ordered observation. Positions 3 through 8 fall in the 10–19 group, but their individual values are not supplied.

222. HSPT-M13-048 - Answer A: 1/4
The favorable region is the triangle below $y = x/2$, with base 1 and height 1/2. Its area is $(1/2)(1)(1/2) = 1/4$ of the unit square.

223. HSPT-M13-049 - Answer C: 5/8
The mixed number is 8/5. Its reciprocal is 5/8 because $(8/5)(5/8) = 1$; merely reversing the fractional part would be incorrect.

224. HSPT-M13-050 - Answer A: 71
Twenty digits contain six full 416 blocks and the next digits 4 and 1. Their sum is $6(4 + 1 + 6) + 4 + 1 = 71$.

225. HSPT-M13-051 - Answer A: 17
17 cannot equal $4a + 7b$ for nonnegative integers. Lengths 18, 19, 20, and 21 are possible; adding 4 covers every larger length, so 17 is the largest gap.

226. HSPT-M13-052 - Answer C: 3
The terms are $3x^2$, $-4x$, and 7; the exponent does not create an extra term, and the constant counts as a term.

227. HSPT-M13-053 - Answer D: $\sqrt{(p/q)}$
Divide by positive q to get $r^2 = p/q$. Since r is positive, take the positive square root: $r = \sqrt{(p/q)}$.

228. HSPT-M13-054 - Answer C: −3 and 3
The denominator must not be zero. Solving $x^2 - 9 = 0$ gives $x = -3$ or $x = 3$, so both inputs are excluded.

229. HSPT-M13-055 - Answer D: $-3 < x < 2$
A negative product requires opposite signs. Between −3 and 2, $x + 3$ is positive and $x - 2$ is negative; outside that interval their signs match.

230. HSPT-M13-056 - Answer A: It is concave.
An interior angle greater than 180° creates an inward indentation, making a noncrossing polygon concave rather than convex.

231. HSPT-M13-057 - Answer B: 20°
The minute hand is 120° clockwise from 12. The hour hand is $90° + (20/60) \times 30° = 100°$, leaving a 20° angle.

232. HSPT-M13-058 - Answer C: $16 + 12\sqrt{2}$ units
Four original side portions remain, each $10 - 3 - 3 = 4$. Each of four cuts has length $\sqrt{(3^2 + 3^2)} = 3\sqrt{2}$, giving $16 + 12\sqrt{2}$.

233. HSPT-M13-059 - Answer A: 3 meters + 40 centimeters
Meters and centimeters both measure length; converting 40 centimeters to 0.4 meter gives a valid total length of 3.4 meters.

234. HSPT-M13-060 - Answer D: 1.8 liters
The cross-sectional area is $(1/2)(6)(3) = 9$ square centimeters. Multiplying by 200 centimeters gives 1,800 cubic centimeters, or 1.8 liters.

235. HSPT-M13-061 - Answer D: It can be 28, 29, 30, or 31 days.
Calendar months have different lengths, including February's year-dependent length; one calendar month therefore cannot be replaced by one fixed number of days.

236. HSPT-M13-062 - Answer B: 0.65
A probability must lie between 0 and 1, inclusive. The value 0.65 meets that bound, while the other listed values do not.

237. HSPT-M13-063 - Answer B: 360
The marked fraction is 6/30 = 1/5. Under the stated model, 72 is one-fifth of the population, so the estimated size is $72 \times 5 = 360$.

238. HSPT-M13-064 - Answer B: 2/3
Of 36 pairs, the 9 odd-odd pairs are excluded. Of the 27 remaining pairs, 18 have mixed parity and an odd sum, giving $18/27 = 2/3$.

Language

239. HSPT-L13-001 - Answer A: We packed nails, screws, etc.
The period belonging to etc. also ends the sentence; adding another period or a terminal comma or semicolon is unnecessary.

240. HSPT-L13-002 - Answer B: One drawer contains spoons; the other, forks.
The comma after other marks where contains is understood, preserving the parallel relationship between the two drawers and their contents.

241. HSPT-L13-003 - Answer C: The package arrived (who sent it?) before daybreak.
The parenthetical question takes its own question mark inside the parentheses, while the surrounding declarative sentence ends with a period.

242. HSPT-L13-004 - Answer A: King Alden IV
Roman numerals used as formal numbers in a ruler's name are conventionally uppercase; IV represents the Fourth in this name.

243. HSPT-L13-005 - Answer A: The Self-Guided Journey
In headline-style capitalization, the first word and both major parts of self-guided are capitalized, as is the main noun Journey.

244. HSPT-L13-006 - Answer C: exaggerate
Exaggerate uses two g letters and one r, with the ending ate; the alternatives omit, duplicate, or replace letters.

245. HSPT-L13-007 - Answer A: camouflage
Camouflage includes the sequence ou after cam and ends in flage; the alternatives omit, move, or add vowels within that pattern.

246. HSPT-L13-008 - Answer C: another; other
Another supplies the determiner for singular route; other can modify plural routes, while another cannot directly modify a plural noun.

247. HSPT-L13-009 - Answer D: Dora said she would deliver the film today.
Monday's tomorrow is Tuesday's today; reported will becomes would, and she preserves Dora as the person responsible for the delivery.

248. HSPT-L13-010 - Answer B: are
The real subject is the plural phrase two gaps, which follows there; it requires are rather than a singular verb phrase.

249. HSPT-L13-011 - Answer D: everyday; every day
Everyday is an adjective meaning ordinary or routine; every day is a two-word expression meaning each day, describing frequency.

250. HSPT-L13-012 - Answer C: allusion; illusion
An allusion refers indirectly to another work or event; an illusion creates a misleading appearance, so the two contexts require different words.

251. HSPT-L13-013 - Answer D: How much help does each successive level provide?
Sentences 2–5 describe increasing levels of assistance and the option to stop, directly answering a question about the help each level provides.

252. HSPT-L13-014 - Answer D: Notice the letters at the beginnings of the lines.
The correct hint points to a relevant feature without explaining which initials to select or revealing the city; the alternatives overexplain or provide no useful direction.

253. HSPT-L13-015 - Answer B: Keep the solution under the last flap, which the solver may open when ready.
The last flap keeps the complete answer available while delaying its exposure until the solver chooses; the alternatives sacrifice access or control.

254. HSPT-L13-016 - Answer C: Summer of '84
The apostrophe precedes 84 because the omitted digits 19 come before those digits; no closing apostrophe is needed.

255. HSPT-L13-017 - Answer A: We will re-cover the sofa.
Re-cover means cover again, while recover ordinarily means retrieve or regain; the hyphen distinguishes the intended word from a different existing word.

256. HSPT-L13-018 - Answer D: a translation of the Quran
Quran is the proper name of a religious text and is capitalized; the surrounding common words do not require capitals.

257. HSPT-L13-019 - Answer D: The design dates from the twenty-first century.
An ordinary century designation is a common descriptive phrase, not a proper historical name, so its words remain lowercase.

258. HSPT-L13-020 - Answer A: acquaintance
Acquaintance includes cq, the vowel sequence ai, and the ending ance; the alternatives omit letters or substitute the wrong ending.

259. HSPT-L13-021 - Answer B: cinnamon
Cinnamon uses double n, a single m, and the ending amon; the alternatives omit an n, double m, or change the final vowel.

260. HSPT-L13-022 - Answer B: silhouette
Silhouette preserves h before ou and doubles t before the final e; the alternatives omit or rearrange parts of that spelling.

261. HSPT-L13-023 - Answer D: whether
Whether can introduce alternative possibilities before a to-infinitive; if, that, and unless do not form that construction in this sentence.

262. HSPT-L13-024 - Answer C: What a huge wave that was!
What introduces the noun-phrase exclamation what a huge wave; the following clause keeps ordinary subject-before-verb order.

263. HSPT-L13-025 - Answer A: The crate is so wide that it cannot fit through the hatch.
Too wide to fit expresses excessive width preventing passage; so wide that it cannot fit preserves both that cause and the negative result.

264. HSPT-L13-026 - Answer A: alternately; Alternatively
Alternately describes taking turns; alternatively introduces another available choice, here leaving both lamps off.

265. HSPT-L13-027 - Answer D: respectively; respectfully
Respectively matches Ari with silver and Bo with bronze in the stated order; respectfully describes the courteous manner of their thanks.

266. HSPT-L13-028 - Answer B: The interviewer asked about a larger handle, and the designer explained that ridges had also been added.
The interviewer asks about size in sentence 3; the designer adds the second change in sentence 4, so both attribution and correction must remain.

267. HSPT-L13-029 - Answer C: It prevents a mistaken claim in the interviewer's question from appearing to be the interview's accepted conclusion.
Sentence 6 explicitly rejects the size-alone conclusion; removing it would leave the interviewer's mistaken interpretation without the designer's essential correction.

268. HSPT-L13-030 - Answer B: After testing ridges at one fixed size, how would you check whether they help at other sizes?
The proposed comparison holds one size fixed; asking about other sizes examines the limits of that future result without inventing completed tests or unsupported findings.

269. HSPT-L13-031 - Answer D: short- and long-term effects
The suspended hyphen after short stands for the shared ending term; long-term retains its complete hyphenated form before effects.

270. HSPT-L13-032 - Answer B: "Wait, I haven't fin—"
An em dash marks an abrupt cutoff, including an unfinished word; an ellipsis normally suggests trailing off rather than sudden interruption.

271. HSPT-L13-033 - Answer A: The gauge fell silent. (Its battery had failed.)
A complete sentence enclosed in parentheses takes its ending period inside the closing parenthesis, with no additional period outside.

272. HSPT-L13-034 - Answer C: I asked Mom whether my dad was home.
Mom functions as a personal name and is capitalized; dad follows my and functions as an ordinary family-relationship noun.

273. HSPT-L13-035 - Answer C: Lakes Vera and Tano
The specified style capitalizes shared Lakes and the proper names Vera and Tano; the linking conjunction and remains lowercase.

274. HSPT-L13-036 - Answer A: brochure
Brochure retains the ch spelling and the ending ure; the alternatives substitute a sound-based spelling or omit a required letter.

275. HSPT-L13-037 - Answer B: mayonnaise
Mayonnaise has double n and the ending aise; the alternatives lose an n, add an s, or insert an extra i.

276. HSPT-L13-038 - Answer B: wait; to wait
Active made takes an object followed by a bare infinitive; its passive counterpart were made takes the to-infinitive to wait.

277. HSPT-L13-039 - Answer A: will have finished
Will have finished is future perfect, placing the expected completion before the later future reference point rather than in the present or past.

278. HSPT-L13-040 - Answer C: She put them away but could not put up with it.
A pronoun separates put from away, but it follows the complete inseparable expression put up with; the two verb patterns require different positions.

279. HSPT-L13-041 - Answer B: defuse
Defuse means remove tension or potential conflict; diffuse means spread out, so it does not express the intended calming action.

280. HSPT-L13-042 - Answer C: loathe; loath
Loathe is the verb meaning intensely dislike; loath is the adjective meaning reluctant, used after am before the infinitive to complain.

281. HSPT-L13-043 - Answer D: historic; historical
Historic emphasizes historical importance; historical describes a connection with past events without necessarily claiming that importance.

282. HSPT-L13-044 - Answer D: Let the lower edge stay unfinished, as though the cloud is still arriving.
The revision repeats Let and preserves the unfinished edge and arriving-cloud image; the alternatives change the proposed action or its meaning.

283. HSPT-L13-045 - Answer A: Leave a little room for the cloud to become something you had not planned.
The open-ended closing extends the arriving-cloud image and the draft's invitation to notice differences; the original sentence imposes the uniformity sentence 5 rejects.

284. HSPT-L13-046 - Answer A: Stone and Water: A Study of Walls
A colon separates the complete main title from its subtitle, so it belongs after Water rather than inside either title element.

285. HSPT-L13-047 - Answer B: Was the parcel sent from the U.S.?
The abbreviation keeps its final period, and the question adds a question mark after it; the two marks serve separate functions.

286. HSPT-L13-048 - Answer A: German shepherd
German is a proper geographic adjective and takes a capital; shepherd remains a common breed noun in the specified form.

287. HSPT-L13-049 - Answer C: Things We Can Agree On
Headline style capitalizes major words and the final word, so On remains capitalized even though a short preposition would ordinarily be lowercase within a title.

288. HSPT-L13-050 - Answer C: peculiar
Peculiar uses one c and the ending uliar; the alternatives reverse vowels, replace the ending, or double the c.

289. HSPT-L13-051 - Answer B: mosquito
Mosquito uses qu, the vowel i, and a single t before the final o; the alternatives alter those features.

290. HSPT-L13-052 - Answer A: entrepreneur
Entrepreneur preserves the opening entre and the ending preneur; the alternatives transpose, omit, or relocate letters within those sequences.

291. HSPT-L13-053 - Answer C: because of; because
Because of introduces the noun fog; because introduces the full clause the visibility worsened, so the two complements require different forms.

292. HSPT-L13-054 - Answer D: where
Where supplies the location relationship for the complete clause the bats spend the winter; which would require an accompanying preposition in this construction.

293. HSPT-L13-055 - Answer A: had seen; would have closed
The unreal past condition takes had seen, while its unreal past result takes would have closed; both refer to the missed opportunity yesterday.

294. HSPT-L13-056 - Answer C: cite
Cite means refer to or acknowledge a source; site names a location, while sight concerns seeing, so neither expresses the reporting action.

295. HSPT-L13-057 - Answer D: council; counsel
Council names a governing body; counsel means advice, so the elected group is the council and what it seeks is counsel.

296. HSPT-L13-058 - Answer B: personnel; personal
Personnel refers to an organization's staff, fitting the officer's administrative role; personal describes private diaries belonging to individual workers.

297. HSPT-L13-059 - Answer D: Immediately after sentence 1
Sentence 1 introduces the reference event; moving its definition directly afterward gives that moment a clear referent before offsets and their signs are explained.

298. HSPT-L13-060 - Answer D: A +3 offset places the sound three frames after the chosen contact frame, wherever that frame occurs in the film.
The offset measures a difference from the selected contact event, not an absolute frame number or a number of seconds from the film's beginning.